Real Analysis Notes

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1 Cardinality

1.1 Basic concepts

Definition 1.1 – Equipotence

Two sets A and B are **equipotent**, or have the same cardinality ($A \sim B$), if there is a bijection $f : A \to B$. It is an equivalence relation: reflexive via the identity, symmetric via the inverse and transitive via composition.

Definition 1.2 – Countable Set

A is **countable** if $A \sim \mathbb{N}$.

Example 1.1 – Integers

$\mathbb{Z}$ is countable: the bijection $f : \mathbb{N} \to \mathbb{Z}$ defined by

$$f(k) = \begin{cases} k/2 & \text{if } k \text{ is even,} \\ -(k-1)/2 & \text{if } k \text{ is odd} \end{cases}$$

shows that $\mathbb{N} \sim \mathbb{Z}$, even though $\mathbb{N} \subsetneq \mathbb{Z}$.

1.2 The Cantor-Bernstein theorem

Theorem 1.1 – Cantor-Bernstein

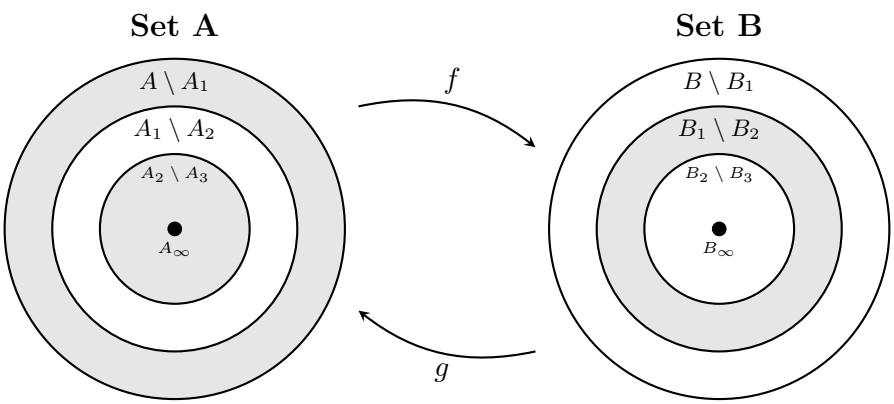
If $f : A \to B$ and $g : B \to A$ are injective, there is a bijection $h : A \to B$. In particular, $A \sim B$.

The idea is to decompose A and B into nested layers and alternate the map according to the layer: on some, f is used, and on others g^{-1}. The set A is split into three classes:

(i) Even layers (gray in the figure): elements of A that do not come from B via g. On them f is used.

(ii) Odd layers (white in the figure): elements that do come from B via g. On them g^{-1} is used.

(iii) The core A_∞: the elements that belong to every layer. On it f is used.



Proof. Define recursively

$$\begin{aligned} A_1 &= g(B) & B_1 &= f(A) \\ A_2 &= g(B_1) & B_2 &= f(A_1) \\ &\vdots & &\vdots \\ A_{k+1} &= g(B_k) & B_{k+1} &= f(A_k) \\ A_\infty &= \bigcap_{k=1}^{\infty} A_k & B_\infty &= \bigcap_{k=1}^{\infty} B_k. \end{aligned}$$

A and B are written as disjoint unions of the layers and the cores, and h acts differently on each part.

$$\begin{aligned} A \;=\; & \underbrace{[(A \setminus A_1) \cup (A_2 \setminus A_3) \cup \dots]}_{\text{Even layers of A}} \;\cup\; \underbrace{[(A_1 \setminus A_2) \cup (A_3 \setminus A_4) \cup \dots]}_{\text{Odd layers of A}} \;\cup\; \underbrace{A_\infty}_{\text{Core of A}} \\ & \qquad\qquad \downarrow f \quad \uparrow f^{-1} \qquad\qquad \downarrow g^{-1} \quad \uparrow g \qquad\qquad \downarrow f \quad \uparrow f^{-1} \\ B \;=\; & \underbrace{[(B_1 \setminus B_2) \cup (B_3 \setminus B_4) \cup \dots]}_{\text{Odd layers of B}} \;\cup\; \underbrace{[(B \setminus B_1) \cup (B_2 \setminus B_3) \cup \dots]}_{\text{Even layers of B}} \;\cup\; \underbrace{B_\infty}_{\text{Core of B}} \end{aligned}$$

Define $h : A \to B$ by cases:

$$h(x) = \begin{cases} f(x) & \text{if } x \in A_k \setminus A_{k+1} \text{ with } k \text{ even}, \\ g^{-1}(x) & \text{if } x \in A_k \setminus A_{k+1} \text{ with } k \text{ odd}, \\ f(x) & \text{if } x \in A_\infty. \end{cases}$$

Injectivity follows from the injectivity of f and g. It remains to check surjectivity on each part.

(i) *Even rings.* The restriction $f : (A_{2k} \setminus A_{2k+1}) \to (B_{2k+1} \setminus B_{2k+2})$ is onto. If $y \in B_{2k+1} \setminus B_{2k+2}$, since $B_{2k+1} = f(A_{2k})$ there is $x \in A_{2k}$ with $f(x) = y$. If $x \in A_{2k+1}$, then $y = f(x) \in f(A_{2k+1}) = B_{2k+2}$, a contradiction. Hence $x \in A_{2k} \setminus A_{2k+1}$.

(ii) *Odd rings.* Now the restriction $g^{-1} : (A_{2k-1} \setminus A_{2k}) \to (B_{2k-2} \setminus B_{2k-1})$. It is equivalent to showing that

$$g : (B_{2k-2} \setminus B_{2k-1}) \to (A_{2k-1} \setminus A_{2k})$$

is a bijection. Take $x \in A_{2k-1} \setminus A_{2k}$; since $A_{2k-1} = g(B_{2k-2})$, there is $y \in B_{2k-2}$ with $g(y) = x$. If $y \in B_{2k-1}$, then $x = g(y) \in g(B_{2k-1}) = A_{2k}$, a contradiction.

(iii) *The core.* For $f : A_\infty \to B_\infty$, given $z \in B_\infty$, $z \in B_{k+1} = f(A_k)$ for every k. There are then $x_k \in A_k$ with $f(x_k) = z$; by injectivity of f it is *the same* x for every k. Hence $x \in \bigcap A_k = A_\infty$ and $z = f(x)$.

h is a bijection and therefore $A \sim B$. □

1.3 Algebra of Countable Sets

Theorem 1.2 – Cartesian Product

The finite Cartesian product of countable sets is countable. In particular, $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$.

Proof. By induction on the number k of factors. The case $k = 1$ is the hypothesis: a single factor, already countable. For the inductive step, by associativity

$$A_1 \times \cdots \times A_n \times A_{n+1} = (A_1 \times \cdots \times A_n) \times A_{n+1} = P_n \times A_{n+1},$$

and everything reduces to showing that the product of two countable sets is countable. That in turn follows from $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$: the function $f(n, m) = 2^n 3^m$ is injective by the Fundamental Theorem of Arithmetic. □

Theorem 1.3 – Countable Union

$$A = \bigcup_{n \in \mathbb{N}} A_n, \quad \text{with } A_n \text{ countable} \implies A \text{ is countable.}$$

Proof. If the A_n are not disjoint, an element may be counted several times, which prevents building a direct injection. Replace (A_n) by a disjoint family with the same union:

$$A'_1 = A_1, \qquad A'_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k \quad (n > 1).$$

Each $A'_n \subseteq A_n$ is countable, the A'_n are pairwise disjoint, and $\bigcup A'_n = \bigcup A_n$.

Hence there are injections $g_n : A'_n \to \mathbb{N}$. Define $H : A \to \mathbb{N} \times \mathbb{N}$: given $x \in A$, there is a unique n with $x \in A'_n$ (because they are disjoint), and set $H(x) = (n, g_n(x))$. H is injective. Since $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$, A is countable. □

Theorem 1.4 – Rationals

$\mathbb{Q}$ is countable.

Proof. By means of the assignment $p/q \mapsto (p, q)$ (with $q \in \mathbb{N}$, $p \in \mathbb{Z}$ and $\gcd(p, q) = 1$), $\mathbb{Q}$ injects into $\mathbb{Z} \times \mathbb{N}$, which is countable. □

1.4 Uncountable Sets

Theorem 1.5 – Uncountability of the Interval

The interval $(0, 1)$ is not countable.

Proof. By Cantor's diagonal argument. If it were countable, its elements could be listed:

$$\begin{aligned} y_1 &= 0.d_{11}d_{12}d_{13}\ldots \\ y_2 &= 0.d_{21}d_{22}d_{23}\ldots \\ &\vdots \\ y_k &= 0.d_{k1}d_{k2}d_{k3}\ldots \end{aligned}$$

Construct $z = 0.z_1 z_2 z_3 \ldots$ with $z_k \neq d_{kk}$ (for example, $z_k = 2$ if $d_{kk} = 1$ and $z_k = 1$ otherwise). Then z differs from each y_k in the k-th digit, so that $z \notin \{y_k\}$, contrary to the assumption that the list covered $(0, 1)$. □

Corollary 1.1 – Reals

$\mathbb{R}$ is not countable.

Proof. First, $(0,1) \sim (a,b)$ via the linear map $t \mapsto a + t(b-a)$. Second, $\mathbb{R} \sim (-\pi/2, \pi/2)$ via arctan. Composition gives $\mathbb{R} \sim (0,1)$, which is not countable. □

A remark on notation and the order of cardinals

Lemma 1 (Injectivity and Surjectivity by Composition)**.** *If $f : A \to B$, $g : B \to A$ and $g \circ f = \mathrm{id}_A$, then f is injective and g is surjective.*

Proof. *f injective.* If $f(x_1) = f(x_2)$, applying g gives $x_1 = g(f(x_1)) = g(f(x_2)) = x_2$.

g surjective. Given $a \in A$, take $b = f(a)$; then $g(b) = a$. □

With this, the order of cardinals is defined:

Definition 1.3 – Inequality of Cardinals

$\#A \leq \#B$ if there is an injection $f : A \to B$.

There is an equivalent formulation with surjections in the other direction:

Proposition 1.1 – Equivalent Form of $\#A \leq \#B$

If $A \neq \emptyset$, $\#A \leq \#B$ if and only if there is a surjection $g : B \to A$.

Proof. Both implications are proved.

($\Rightarrow$). Let $f : A \to B$ be injective. Define $g : B \to A$: for $b \in f(A)$, let $g(b)$ be the unique a with $f(a) = b$, and for $b \notin f(A)$ assign a fixed $a_0 \in A$ (here $A \neq \emptyset$ is used). Thus $g \circ f = \mathrm{id}_A$ and, by Lemma 1, g is surjective.

($\Leftarrow$). Let g be surjective. By the axiom of choice, for each a choose $b_a \in g^{-1}(\{a\})$ and define $f(a) = b_a$. Then $g \circ f = \mathrm{id}_A$ and f is injective. □

1.5 Cantor's Theorem (Power Set)

Theorem 1.6 – Cantor's Theorem

$|A| < |\mathcal{P}(A)|$ for every set A.

Proof. $|A| \leq |\mathcal{P}(A)|$ because $a \mapsto \{a\}$ is injective. To see that there is no surjection $h : A \to \mathcal{P}(A)$, argue by contradiction (diagonal argument): if h were onto, with the diagonal set

$$B = \{x \in A : x \notin h(x)\} \in \mathcal{P}(A)$$

there would be $b \in A$ with $h(b) = B$. But

- $b \in B \implies b \notin h(b) = B$,
- $b \notin B \implies b \in h(b) = B$.

Both are absurd, so that no such h exists. □

1.6 Solved exercises

Exercise 1

Let A and B be subsets of X.

Proposition 1.2 – Exercise 1.i

Prove that $A \subset B \iff B^c \subset A^c$.

Proof. Both implications are proved.

($\Rightarrow$). Suppose that $A \subset B$; it must be shown that $B^c \subset A^c$.

- Let $x \in B^c$, that is, $x \notin B$.
- By hypothesis $A \subset B$, so that if x were in A it would be in B.
- Since $x \notin B$, by contraposition $x \notin A$.
- If $x \notin A$, then $x \in A^c$.

($\Leftarrow$). Suppose that $B^c \subset A^c$; it must be shown that $A \subset B$.

- Let $x \in A$, that is, $x \notin A^c$.
- By hypothesis ($B^c \subset A^c$), if x belonged to B^c it would belong to A^c.
- Since $x \notin A^c$, it cannot be that $x \in B^c$.
- Therefore $x \notin B^c$, which implies $x \in B$.

□

Proposition 1.3 – Exercise 1.ii

Prove that $B - \bigcup_i A_i = \bigcap_i (B - A_i)$.

Proof. Double inclusion is proved.

($\subseteq$). Let $x \in B - \bigcup_i A_i$.

- By definition of difference, $x \in B$ *and* $x \notin \bigcup_i A_i$.
- That x is not in the union means that it belongs to none of the A_i; that is, $\forall i, x \notin A_i$.
- Since $x \in B$ and moreover $\forall i, x \notin A_i$, $\forall i, (x \in B \wedge x \notin A_i)$.
- This is equivalent to $\forall i, x \in (B - A_i)$.
- By definition of generalized intersection, $x \in \bigcap_i (B - A_i)$.

($\supseteq$). Let $x \in \bigcap_i (B - A_i)$.

- By definition of intersection, $\forall i, x \in (B - A_i)$.
- That is, for every i: $x \in B$ *and* $x \notin A_i$.
- Two things follow from this:

  (i) $x \in B$ (common to all).

(ii) $\forall i, x \notin A_i$.

- The second condition implies that there is no index with $x \in A_i$; therefore $x \notin \bigcup_i A_i$.
- Putting together $x \in B$ and $x \notin \bigcup_i A_i$, one obtains $x \in B - \bigcup_i A_i$.

□

Proposition 1.4 – Exercise 1.iii

Prove that $B - \bigcap_i A_i = \bigcup_i (B - A_i)$.

Proof. Double inclusion is proved.

($\subseteq$). Let $x \in B - \bigcap_i A_i$.

- Then $x \in B$ *and* $x \notin \bigcap_i A_i$.
- Denying that x is in the intersection means that it fails for at least one: there is some index k with $x \notin A_k$.
- Since $x \in B$ and for that k one has $x \notin A_k$, then $x \in (B - A_k)$.
- Since x belongs to one of the sets of the family, it belongs to the union.

($\supseteq$). Let $x \in \bigcup_i (B - A_i)$.

- By definition of union, there is some index k with $x \in (B - A_k)$.
- That is, $x \in B$ and $x \notin A_k$.
- Since $x \notin A_k$, x cannot be in the intersection of all the A_i (since $x \notin A_k$). Therefore $x \notin \bigcap_i A_i$.
- Having $x \in B$ and $x \notin \bigcap_i A_i$, it follows that $x \in B - \bigcap_i A_i$.

□

Proposition 1.5 – Exercise 1.iv

Prove that $\bigcup_i (A_i \cap B) = (\bigcup_i A_i) \cap B$.

Proof. Double inclusion is proved.

($\subseteq$). Let $x \in \bigcup_i (A_i \cap B)$.

- There is some index k with $x \in (A_k \cap B)$.
- That is, $x \in A_k$ and $x \in B$.
- Since $x \in A_k$, then $x \in \bigcup_i A_i$ (being in one suffices).
- Since $x \in \bigcup_i A_i$ and moreover $x \in B$, then $x \in (\bigcup_i A_i) \cap B$.

($\supseteq$). Let $x \in (\bigcup_i A_i) \cap B$.

- Then $x \in \bigcup_i A_i$ and $x \in B$.
- Being in the union, there is an index k with $x \in A_k$.
- Since $x \in A_k$ and $x \in B$, then $x \in (A_k \cap B)$.

- Belonging to one of the terms $(A_k \cap B)$, it belongs to the union.

□

Exercise 2

Given a sequence $(A_n)_{n\in\mathbb{N}}$ of sets, and $A = \bigcup_n A_n$, find a sequence $(B_n)_{n\in\mathbb{N}}$ of disjoint sets such that $A = \bigcup_n B_n$.

Proof. Define the sequence (B_n) recursively:

$$B_1 = A_1$$
$$B_n = A_n \setminus \left(\bigcup_{k=1}^{n-1} A_k \right) \quad \text{for } n > 1$$

Two things must be proved: that they are disjoint and that their union is A.

Disjoint. Let $n, m \in \mathbb{N}$ with $n \neq m$; without loss of generality, $m < n$. By definition,

$$B_n = A_n \setminus (A_1 \cup \cdots \cup A_m \cup \cdots \cup A_{n-1}),$$

so that B_n has no element of A_m, that is, $B_n \cap A_m = \emptyset$. Since $B_m \subseteq A_m$ by construction, one has $B_n \cap B_m = \emptyset$.

Equality of the unions. Double inclusion is proved.

($\subseteq$) Since $B_n \subseteq A_n$ for every n,

$$\bigcup_{n=1}^{\infty} B_n \subseteq \bigcup_{n=1}^{\infty} A_n = A.$$

($\supseteq$) Let $x \in A = \bigcup_{n=1}^{\infty} A_n$; then x belongs to at least one of the A_n. Let k be the least index with $x \in A_k$. This implies $x \in A_k$ and $x \notin A_j$ for every $j < k$ (otherwise, k would not be the least). Hence x does not belong to the union of the previous ones:

$$x \notin \bigcup_{j=1}^{k-1} A_j,$$

and by definition of B_k one has $x \in A_k \setminus \bigcup_{j=1}^{k-1} A_j$, that is, $x \in B_k$. Therefore $x \in \bigcup_{n=1}^{\infty} B_n$.

Finally, $A = \bigcup_n B_n$. □

Exercise 3

Given a function $f : X \longrightarrow Y$ and subsets A, B of X and C, D of Y.

Proposition 1.6 – 3.i

$f(A \cup B) = f(A) \cup f(B)$

Proof. Double inclusion is proved.

($\subseteq$). Let $y \in f(A \cup B)$. By definition of direct image, there is $x \in A \cup B$ with $f(x) = y$. Since $x \in A$ or $x \in B$:

- if $x \in A$, then $f(x) \in f(A)$, hence $y \in f(A)$;
- if $x \in B$, then $f(x) \in f(B)$, hence $y \in f(B)$.

In either case, $y \in f(A) \cup f(B)$.

($\supseteq$). Let $y \in f(A) \cup f(B)$, that is, $y \in f(A)$ or $y \in f(B)$.

- If $y \in f(A)$, there is $x \in A$ with $f(x) = y$. Since $A \subset A \cup B$, then $x \in A \cup B$, so that $y \in f(A \cup B)$.
- Analogously, if $y \in f(B)$, there is $x \in B$ with $f(x) = y$, and $y \in f(A \cup B)$.

□

Proposition 1.7 – 3.ii

$f(A \cap B) \subset f(A) \cap f(B)$. Does equality hold?

Proof. Let $y \in f(A \cap B)$. There is $x \in A \cap B$ with $f(x) = y$. Since $x \in A$ and $x \in B$, one has $f(x) \in f(A)$ and $f(x) \in f(B)$, hence $y \in f(A) \cap f(B)$. □

Equality does not hold in general. As a counterexample, let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, and let $A = [-1, 0]$, $B = [0, 1]$. Then $A \cap B = \{0\}$, so that $f(A \cap B) = \{0\}$; on the other hand $f(A) = [0, 1] = f(B)$, so that $f(A) \cap f(B) = [0, 1] \neq \{0\}$.

Proposition 1.8 – 3.iii

$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$

Proof. Let x be an arbitrary element:

$$\begin{aligned} x \in f^{-1}(C \cup D) &\iff f(x) \in C \cup D \\ &\iff f(x) \in C \vee f(x) \in D \\ &\iff x \in f^{-1}(C) \vee x \in f^{-1}(D) \\ &\iff x \in f^{-1}(C) \cup f^{-1}(D). \end{aligned}$$

□

Proposition 1.9 – 3.iv

$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$

Proof. Let x be an arbitrary element:

$$\begin{aligned} x \in f^{-1}(C \cap D) &\iff f(x) \in C \cap D \\ &\iff f(x) \in C \wedge f(x) \in D \\ &\iff x \in f^{-1}(C) \wedge x \in f^{-1}(D) \\ &\iff x \in f^{-1}(C) \cap f^{-1}(D). \end{aligned}$$

□

Proposition 1.10 – 3.v

$A \subset f^{-1}(f(A))$. Does equality hold?

Proof. Let $x \in A$. Applying f, $f(x) \in f(A)$. By definition of preimage, $f^{-1}(f(A))$ contains all the elements whose image lies in $f(A)$; since $f(x) \in f(A)$, then $x \in f^{-1}(f(A))$. □

Equality does not hold in general (only if f is injective). As a counterexample, let $f(x) = x^2$ and $A = \{1\}$: then $f(A) = \{1\}$ and $f^{-1}(f(A)) = f^{-1}(\{1\}) = \{-1, 1\}$, with $\{1\} \subsetneq \{-1, 1\}$.

Proposition 1.11 – 3.vi

$f(f^{-1}(C)) \subset C$. Does equality hold?

Proof. Let $y \in f(f^{-1}(C))$. By definition of direct image, there is $x \in f^{-1}(C)$ with $f(x) = y$; and by definition of preimage, $x \in f^{-1}(C)$ implies $f(x) \in C$. Since $f(x) = y$, it follows that $y \in C$. □

Equality does not hold in general (only if f is surjective or if $C \subset \mathrm{Im}(f)$). As a counterexample, let $f(x) = e^x$ from $\mathbb{R}$ to $\mathbb{R}$ and $C = \mathbb{R}$: then $f^{-1}(C) = \mathbb{R}$ and $f(f^{-1}(C)) = f(\mathbb{R}) = (0, \infty)$, with $(0, \infty) \subsetneq \mathbb{R}$.

Proposition 1.12 – 3.vii

$f^{-1}(D^c) = (f^{-1}(D))^c$

Proof. Let x be an arbitrary element:

$$\begin{aligned} x \in f^{-1}(D^c) &\iff f(x) \in D^c \\ &\iff f(x) \notin D \\ &\iff \neg(f(x) \in D) \\ &\iff \neg(x \in f^{-1}(D)) \\ &\iff x \notin f^{-1}(D) \\ &\iff x \in (f^{-1}(D))^c. \end{aligned}$$

□

Exercise 4

Proposition 1.13 – Relation → Partition

Let X be a set and $\sim$ an equivalence relation on X. The set of equivalence classes (quotient set) $X/\sim = \{[x] : x \in X\}$ forms a partition of X.

Proof. A family of sets is a partition if it satisfies three conditions.

The classes are nonempty. Let $[x]$ be an arbitrary class. By reflexivity, $x \sim x$, and by definition of class ($[x] = \{y \in X : y \sim x\}$) this gives $x \in [x]$. Hence $[x] \neq \emptyset$.

The union covers X. The union of subsets of X is contained in X. For the other inclusion, let $x \in X$; since $x \in [x]$, the element x belongs to the union of all the classes. Therefore

$$X = \bigcup_{x \in X} [x].$$

They are pairwise disjoint. Let $[x]$ and $[y]$ be two classes; it must be shown that they are disjoint or identical. Suppose that they are not disjoint, that is, there is $z \in [x] \cap [y]$.

- $z \in [x] \implies z \sim x \implies x \sim z$.
- $z \in [y] \implies z \sim y$.
- By transitivity, $x \sim y$.

Now it is shown that $[x] = [y]$. Let $w \in [x]$, that is $w \sim x$; since $w \sim x$ and $x \sim y$, by transitivity $w \sim y$, hence $w \in [y]$. This proves $[x] \subseteq [y]$, and by symmetry the opposite inclusion holds. Thus, two classes that share an element are equal; if they are distinct, their intersection is empty. □

Proposition 1.14 – Partition → Relation

Let $P = \{A_i\}_{i \in I}$ be a partition of a set X. The relation defined by $x \sim y \iff \exists k \in I$ such that $x \in A_k \wedge y \in A_k$ is an equivalence relation.

Proof. The three properties are verified.

Reflexivity. Let $x \in X$. Since P is a partition, the union of the A_i covers X, so that there is an index k with $x \in A_k$. Then x and x are in the same A_k, hence $x \sim x$.

Symmetry. Suppose $x \sim y$, that is, there is k with $x, y \in A_k$. Then also $y, x \in A_k$, hence $y \sim x$.

Transitivity. Suppose $x \sim y$ and $y \sim z$.

- $x \sim y \implies \exists k$ with $x, y \in A_k$.
- $y \sim z \implies \exists j$ with $y, z \in A_j$.

The element y belongs both to A_k and to A_j. Since the elements of P are pairwise disjoint, the only way for $A_k \cap A_j \neq \emptyset$ is $A_k = A_j$. Hence x and z are in the same set, and $x \sim z$. □

Exercise 5

This exercise has already been solved.

Exercise 6

The exercise proposes using the principle of complete induction. Given a property $P(n)$, define the truth set $S = \{n \in \mathbb{N} : P(n) \text{ is true}\}$. If it is proved that $1 \in S$ and that the property holding for all the previous ones implies that it holds for the next one, then $S = \mathbb{N}$ (the formula holds for all of them).

Proposition 1.15 – 6.i

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

Proof. By induction on n. Let $S = \{n \in \mathbb{N} : \sum_{k=1}^{n} k = \frac{n(n+1)}{2}\}$.

Base case ($n = 1$). The left-hand side is $\sum_{k=1}^{1} k = 1$ and the right-hand side is $\frac{1(1+1)}{2} = \frac{2}{2} = 1$. Since they coincide, $1 \in S$.

Inductive step. Suppose that $n \in S$; it must be proved that $\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}$. Starting from the sum up to $n+1$,

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^{n} k\right) + (n+1).$$

Using the inductive hypothesis on the sum up to n,

$$= \frac{n(n+1)}{2} + (n+1).$$

Factoring out $(n+1)$ and simplifying,

$$= (n+1)\left(\frac{n}{2} + 1\right) = (n+1)\left(\frac{n+2}{2}\right) = \frac{(n+1)(n+2)}{2}.$$

Hence $n+1 \in S$, and by induction $S = \mathbb{N}$. □

Proposition 1.16 – 6.ii

$$\sum_{k=0}^{n} r^k = \frac{r^{n+1} - 1}{r - 1} \quad (r \neq 1)$$

Proof. By induction on n. Let $S = \{n \in \mathbb{N} : \sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1}\}$.

Base case ($n = 1$). The left-hand side is $\sum_{k=0}^{1} r^k = r^0 + r^1 = 1 + r$ and the right-hand side $\frac{r^{1+1}-1}{r-1} = \frac{r^2-1}{r-1} = \frac{(r-1)(r+1)}{r-1} = r + 1$. Since they coincide, $1 \in S$.

Inductive step. Suppose that $n \in S$; it must be proved that the sum up to $n+1$ equals $\frac{r^{n+2}-1}{r-1}$. Starting from the sum,

$$\sum_{k=0}^{n+1} r^k = \left(\sum_{k=0}^{n} r^k\right) + r^{n+1},$$

using the inductive hypothesis,

$$= \frac{r^{n+1} - 1}{r - 1} + r^{n+1},$$

and taking a common denominator, distributing and cancelling r^{n+1} with $-r^{n+1}$,

$$= \frac{r^{n+1} - 1 + r^{n+1}(r-1)}{r-1} = \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r-1} = \frac{r^{n+2} - 1}{r-1}.$$

This is the formula at $n+1$, hence $n+1 \in S$ and by induction $S = \mathbb{N}$. □

Exercise 7

Prove the Well-Ordering Principle: every nonempty subset of $\mathbb{N}$ has a minimum.

Proof. Argue by contradiction. Let $A \subseteq \mathbb{N}$ be nonempty and suppose that A has no minimum. Define the set of naturals strictly smaller than every element of A:

$$S = \{n \in \mathbb{N} : \forall a \in A,\ n < a\}.$$

It is proved by induction that $S = \mathbb{N}$.

Base case. Since 1 is the least natural ($1 \leq x$ for every $x \in \mathbb{N}$), if $1 \in A$ then 1 would be the minimum of A. Since A has no minimum, $1 \notin A$; hence $\{1\} \cap A = \emptyset$, which gives $1 \in S$.

Inductive step. Suppose $n \in S$, that is, $\{1, 2, \ldots, n\} \cap A = \emptyset$. If $n+1 \in A$, since $1, \ldots, n$ are not in A, then $n+1$ would be the minimum of A, which does not exist. Therefore $n+1 \notin A$, so that $\{1, 2, \ldots, n, n+1\} \cap A = \emptyset$ and $n+1 \in S$.

By induction, $S = \mathbb{N}$, that is, $n \notin A$ for every natural n, and consequently $A = \emptyset$. This contradicts A being nonempty. The contradiction comes from supposing that A has no minimum; hence A has a minimum. □

Exercise 8

Find the cardinality of the following sets.

- Set: $\mathbb{Z}_{\geq 3} = \{n \in \mathbb{Z} : n \geq 3\}$. Cardinality: $\aleph_0$.
  It is an infinite subset of a countable set ($\mathbb{Z}$), therefore it is countable. The explicit bijection $f : \mathbb{N} \to \mathbb{Z}_{\geq 3}$ is given by $f(n) = n + 2$.
- Set: $5\mathbb{Z} = \{5k : k \in \mathbb{Z}\}$. Cardinality: $\aleph_0$.
  Define the function $f : \mathbb{Z} \to 5\mathbb{Z}$ as $f(k) = 5k$, which is bijective (injective because $5k_1 = 5k_2 \implies k_1 = k_2$ and surjective by definition). Since $\mathbb{Z} \sim \mathbb{N}$, then $5\mathbb{Z} \sim \mathbb{N}$.
- Set: $\mathbb{Z} \times \mathbb{N}$. Cardinality: $\aleph_0$.
  Both $\mathbb{Z}$ and $\mathbb{N}$ are countable. By Theorem 1.2, the finite Cartesian product of countable sets is countable.
- Set: $\mathbb{R} - \mathbb{Q}$. Cardinality: c.
  One has $\mathbb{R} = (\mathbb{R} - \mathbb{Q}) \cup \mathbb{Q}$. By contradiction: if the irrationals were countable, since $\mathbb{Q}$ is countable, $\mathbb{R}$ would be a union of two countable sets, and hence countable; this contradicts the uncountability of the reals. Hence $\mathbb{R} - \mathbb{Q}$ is uncountable.
- Set: $(-1, 1) \cap \mathbb{Q}$. Cardinality: $\aleph_0$.
  It is a subset of $\mathbb{Q}$, and every subset of a countable set is at most countable.

Exercise 9

Prove that the set of prime numbers is countable.

Proof. Let P be the set of prime numbers. Since $P \subset \mathbb{N}$, it suffices to see that it is infinite. Argue by contradiction: suppose that there are finitely many primes $p_1, p_2, \ldots, p_n$, and consider

$$Q = (p_1 \cdot p_2 \cdot \cdots \cdot p_n) + 1.$$

The number Q has some prime divisor q.

- If q were one of the primes in the list, q would divide the product $p_1 \dots p_n$.
- Since q divides Q and the product, it divides the difference $Q - (p_1 \dots p_n) = 1$.
- But a prime cannot divide 1.

This is a contradiction, so that there are infinitely many primes. Being an infinite subset of $\mathbb{N}$, P is countable. □

Exercise 10

Prove that $P(A) \sim \{0,1\}^A$. Become convinced that $\mathbb{R} \sim P(\mathbb{N})$ and that there are more real numbers than naturals.

Proposition 1.17 – Equipotence of Power Set and Characteristic Functions

The power set of A, denoted $P(A)$, is equipotent with the set of functions from A to $\{0,1\}$, denoted $\{0,1\}^A$.

Proof. Define the function $\Phi : P(A) \to \{0,1\}^A$ that assigns to each subset $B \subseteq A$ its characteristic function χ_B, where

$$\chi_B(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \notin B. \end{cases}$$

It is shown that Φ is bijective.

Injectivity. Let $B, C \subseteq A$ with $\Phi(B) = \Phi(C)$, that is, $\chi_B(x) = \chi_C(x)$ for every $x \in A$. Then

$$x \in B \iff \chi_B(x) = 1 \iff \chi_C(x) = 1 \iff x \in C,$$

hence $B = C$.

Surjectivity. Let $f : A \to \{0,1\}$ be an arbitrary function. A set B whose characteristic function is f is sought; define the set of elements that f sends to 1:

$$B = \{x \in A : f(x) = 1\}.$$

Then $B \subseteq A$, so that $B \in P(A)$. By construction, $\chi_B(x) = 1 \iff x \in B \iff f(x) = 1$ and $\chi_B(x) = 0 \iff x \notin B \iff f(x) = 0$, so that $\Phi(B) = f$.

Hence Φ is a bijection and $P(A) \sim \{0,1\}^A$. □

To become convinced that $\mathbb{R} \sim P(\mathbb{N})$:

- It was already proved that $P(\mathbb{N}) \sim \{0,1\}^{\mathbb{N}}$.
- The elements of $\{0,1\}^{\mathbb{N}}$ are infinite sequences of zeros and ones (for example $0, 1, 1, 0, \dots$).
- The real numbers in $(0,1)$ are represented by their binary expansion

$$x = 0.d_1 d_2 d_3 \dots \quad \text{with } d_i \in \{0,1\}.$$

- There is an almost bijective correspondence between these sequences and the reals (except for the cases of double representation, such as $0.100\cdots = 0.011\dots$).
- Therefore the cardinality of $\mathbb{R}$ coincides with that of the binary sequences, which is that of the power set of $\mathbb{N}$.
- Since by Cantor's Theorem 1.6, $|P(\mathbb{N})| > |\mathbb{N}|$, there are strictly more reals than naturals.

Exercise 11

Given $B \subset \mathbb{N}$, prove that if B is equipotent with a proper subset of itself, then B is countable.

Proof. Let $B \subset \mathbb{N}$ be such that there exist $B' \subsetneq B$ and a bijection $f : B \to B'$. Since $B \subseteq \mathbb{N}$, B can only be finite or countable.

Suppose, by contradiction, that B were finite. Two finite sets in bijection have the same cardinality, so that $|B| = |B'|$; but since $B' \subsetneq B$ is a proper subset, $|B'| < |B|$. This gives the contradiction $|B| < |B|$, so that B cannot be finite.

Being an infinite subset of the naturals, B is countable. □

Exercise 12

Given a countable set I, prove that there is an uncountable family $(I_\alpha)_{\alpha \in A}$ of infinite subsets of I such that for each $\alpha \neq \beta$, the intersection $I_\alpha \cap I_\beta$ is finite.

Proof. Since I is countable, it is identified with $\mathbb{Q}$. Take as index set the reals ($A = \mathbb{R}$), which is uncountable.

For each $\alpha \in \mathbb{R}$ choose a sequence of distinct rationals $(x_n)_{n \in \mathbb{N}}$ converging to α, and define

$$I_\alpha = \{x_n : n \in \mathbb{N}\}.$$

Since the terms are distinct, each I_α is infinite.

Let $\alpha, \beta \in \mathbb{R}$ with $\alpha \neq \beta$. By contradiction, suppose that $I_\alpha \cap I_\beta$ is infinite.

- Let $S = I_\alpha \cap I_\beta$. Being infinite, its elements are ordered into a subsequence (y_k).
- Since $S \subset I_\alpha$, the y_k form a subsequence of the one converging to α, hence $y_k \to \alpha$.
- Since $S \subset I_\beta$, the y_k form a subsequence of the one converging to β, hence $y_k \to \beta$.
- By uniqueness of the limit (2.6), $\alpha = \beta$.

This contradicts $\alpha \neq \beta$. Therefore $I_\alpha \cap I_\beta$ is finite. □

2 The real line

2.1 Bounds, Supremum and Maximum

Definition 2.1 – Upper Bound, Supremum and Maximum

Let $A \subseteq \mathbb{R}$.

- **Upper bound:** $c \in \mathbb{R}$ with $a \leq c$ for every $a \in A$. If one exists, A is bounded above.
- **Supremum** $s = \sup(A)$**:** the least upper bound. It requires that s be an upper bound and that every other upper bound c satisfy $s \leq c$.
- **Maximum** $m = \max(A)$**:** a supremum that belongs to the set.

Dually, for lower bounds:

Definition 2.2 – Lower Bound, Infimum and Minimum

Lower bound: $c \leq a$ for every $a \in A$. **Infimum:** the greatest lower bound, $\inf(A)$. **Minimum:** an infimum that belongs to the set.

Remark. One must distinguish between having bounds and having a maximum or minimum.

- There may be no bounds (for example $\mathbb{N}$ is not bounded above), nor supremum, nor maximum.
- If there is an upper bound, there are infinitely many (any larger number is one as well). But if the supremum exists, it is unique.
- Having a maximum is stronger than having a supremum: $\max(A)$, when it exists, coincides with $\sup(A)$. But there may be a supremum without a maximum: $\sup[0, 1) = 1$ and it is not a maximum.

2.2 The Completeness Axiom

Theorem 2.1 – Completeness Axiom

Every nonempty $A \subset \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Remark. In $\mathbb{Q}$ this fails. The set $\{x \in \mathbb{Q} : x^2 < 2\}$ is bounded by 2 but its "edge" $\sqrt{2}$ is not rational.

Proposition 2.1 – Characterization of the Supremum

Let s be an upper bound of A. $s = \sup(A)$ if and only if for every $\epsilon > 0$ there is $a \in A$ with $s - \epsilon < a$.

Proof. Both implications are proved.

($\Rightarrow$) By contradiction: if it did not hold, there would be $\epsilon > 0$ such that $a \leq s - \epsilon$ for every $a \in A$, and then $s - \epsilon$ would be an upper bound strictly smaller than s, a contradiction.

(⇐) Let c be any upper bound; by contradiction, if $c < s$, taking $\epsilon = s - c > 0$ the hypothesis gives $a_0 \in A$ with $a_0 > s - \epsilon = c$, which contradicts c being an upper bound. Thus $s \leq c$ for every upper bound. □

Proposition 2.2 – Monotonicity of the Supremum and Infimum

If $A \subseteq B$ with B bounded, then $\inf(B) \leq \inf(A) \leq \sup(A) \leq \sup(B)$.

Proof. If $a \in A$, then $a \in B$ and $a \leq \sup(B)$, so that $\sup(B)$ is an upper bound of A and $\sup(A) \leq \sup(B)$. The infima are analogous. □

Proposition 2.3 – Archimedean Property

For every $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n > x$. Equivalently, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. Argue by contradiction: if $\mathbb{N}$ were bounded above, by the Completeness Axiom 2.1 it would have a supremum s. Then $s - 1$ is not an upper bound and there is $n \in \mathbb{N}$ with $n > s - 1$, which gives $n + 1 > s$, contradicting s being an upper bound. The equivalent form is obtained by taking $x = 1/\epsilon$. □

Proposition 2.4 – Density of $\mathbb{Q}$ in $\mathbb{R}$

Between two reals $x < y$ there is always a rational.

Proof. By the Archimedean Property 2.3, there is n with $1/n < y - x$, that is $ny - nx > 1$. Between two reals separated by more than 1 there is always an integer m (take m the least integer greater than nx). Then $nx < m < ny$, and dividing, $x < m/n < y$. □

2.3 Sequences and Subsequences

Definition 2.3 – Sequence

A **sequence** of reals is a function $x : \mathbb{N} \to \mathbb{R}$, denoted $(x_n)_{n \in \mathbb{N}}$.

Definition 2.4 – Subsequence

A **subsequence** of (x_n) is $(x_{n_k})_{k \in \mathbb{N}}$, where $k \mapsto n_k$ is strictly increasing.

Example 2.1 – The Subsequence $1/2^k$

From the harmonic sequence $x_n = 1/n$, taking $n_k = 2^k$ yields $x_{n_k} = 1/2^k$: an accelerated version of the original.

2.4 Limit of a Sequence

Definition 2.5 – Limit

(x_n) **converges** to a (written $x_n \to a$ or $\lim x_n = a$) if

$$\forall \epsilon > 0,\ \exists n_\epsilon \in \mathbb{N} :\ n \geq n_\epsilon \implies |x_n - a| < \epsilon.$$

$|x_n - a| < \epsilon$ is equivalent to $x_n \in (a - \epsilon, a + \epsilon)$. However small the neighborhood around a, the sequence eventually enters it and stays: finitely many terms may remain outside (the first $n_\epsilon - 1$), but all the others are inside.

2.5 Boundedness and Uniqueness

Proposition 2.5 – Convergence Implies Boundedness

Every convergent sequence is bounded.

Proof. Let $x_n \to a$. For $\epsilon = 1$ there is n_0 with $|x_n - a| < 1$ for every $n \geq n_0$. The previous ones $\{x_1, \ldots, x_{n_0-1}\}$ are finitely many: taking $M = \max(\{a+1\} \cup \{x_1, \ldots, x_{n_0-1}\})$ and m analogously, $m \leq x_n \leq M$ for every n. □

Proposition 2.6 – Uniqueness of the Limit

If (x_n) converges, the limit is unique.

Proof. Argue by contradiction: suppose $x_n \to a$ and $x_n \to b$ with $a \neq b$, and take $\epsilon = |a-b|/2 > 0$. For large n, $|x_n - a| < \epsilon$ and $|x_n - b| < \epsilon$, and by the triangle inequality

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\epsilon = |a - b|,$$

a contradiction. □

2.6 Algebra of Limits

Theorem 2.2 – Algebra of Sequences

Let (x_n) and (y_n) be sequences such that $x_n \to a$ and $y_n \to b$. Then:

(i) *Sum:* $\lim(x_n + y_n) = a + b$

(ii) *Difference:* $\lim(x_n - y_n) = a - b$

(iii) *Product:* $\lim(x_n \cdot y_n) = a \cdot b$

(iv) *Quotient:* If $b \neq 0$ and $y_n \neq 0$, $\lim\left(\frac{x_n}{y_n}\right) = \frac{a}{b}$

Proof. *Sum.* By the triangle inequality,

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b|.$$

Given $\epsilon > 0$, choose n_1, n_2 that keep each term below $\epsilon/2$; for $n \geq \max\{n_1, n_2\}$ the sum is less than ϵ.

Difference. Same as the sum.

Product. Adding and subtracting $x_n b$,

$$|x_n y_n - ab| \leq |x_n||y_n - b| + |b||x_n - a|.$$

Since (x_n) converges, it is bounded by some K. Choose n_1 with $|y_n - b| < \epsilon/(2K)$ and n_2 with $|x_n - a| < \epsilon/(2|b|)$ (if $b = 0$ the second term becomes small directly). For $n \geq \max\{n_1, n_2\}$ everything adds up to less than ϵ.

Quotient. It suffices to see $1/y_n \to 1/b$ and apply the product. One has

$$\left|\frac{1}{y_n} - \frac{1}{b}\right| = \frac{|y_n - b|}{|y_n||b|}.$$

Since $y_n \to b \neq 0$, there is n_0 with $|y_n| > |b|/2$ for $n \geq n_0$, so that $1/|y_n| < 2/|b|$ and

$$\left|\frac{1}{y_n} - \frac{1}{b}\right| < \frac{2}{|b|^2}|y_n - b|.$$

Choosing $n_1 \geq n_0$ with $|y_n - b| < \epsilon|b|^2/2$ makes it less than ϵ. □

2.7 Preservation of Order

Proposition 2.7 – Preservation of Order

Let $x_n \to a$ and $y_n \to b$. If $x_n \leq y_n$ for every n, then $a \leq b$.

Proof. Argue by contradiction: suppose $a > b$. Taking $\epsilon = (a - b)/2$, there are disjoint neighborhoods around each one: $b + \epsilon = a - \epsilon = (a + b)/2$. For large n, $x_n > a - \epsilon$ and $y_n < b + \epsilon$, so that $y_n < (a + b)/2 < x_n$, contradicting $x_n \leq y_n$. □

2.8 Monotone Convergence Theorem

Theorem 2.3 – Increasing Bounded Sequences

If (x_n) is increasing and bounded above, it converges.

Proof. The set $A = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above, so that by completeness it has a supremum s. It is shown that $x_n \to s$. Given $\epsilon > 0$, $s - \epsilon$ is not an upper bound, so that there is $x_{n_\epsilon} > s - \epsilon$. Since (x_n) is increasing and s is an upper bound, for $n \geq n_\epsilon$ one gets

$$s - \epsilon < x_{n_\epsilon} \leq x_n \leq s,$$

that is, $|x_n - s| < \epsilon$. □

The same holds for decreasing sequences bounded below: they converge to the infimum.

2.9 Limit Points

Definition 2.6 – Limit Point

$a \in \mathbb{R}$ is a **limit point** of (x_n) if for every $\epsilon > 0$ and every $k \in \mathbb{N}$ there is $n > k$ with $|x_n - a| < \epsilon$.

The difference with the limit lies in permanence: the limit requires the sequence to enter the neighborhood and stay in it; a limit point only requires it to enter infinitely often, even if it may leave and enter again.

Example 2.2 – Visual Difference

$x_n = (-1)^n$ has no limit (it oscillates), but it has two limit points: 1 and -1. However large N is, there is an even $n > N$ with $x_n = 1$ (and analogously for -1).

2.10 Characterization of Limit Points

Theorem 2.4 – Characterization by Subsequences

x is a limit point of (x_n) if and only if there is a subsequence $(x_{n_k}) \to x$.

Proof. Both implications are proved.

($\Rightarrow$) Construct n_k one by one with radii $\epsilon = 1/k$: for $k = 1$ take any n_1 with $|x_{n_1} - x| < 1$; for $k > 1$, with $n_1 < \cdots < n_{k-1}$ already chosen, by definition of limit point there is $n_k > n_{k-1}$ with $|x_{n_k} - x| < 1/k$. Given $\epsilon > 0$, choosing k with $1/k < \epsilon$ yields $|x_{n_k} - x| < \epsilon$.

($\Leftarrow$) Let $(x_{n_k}) \to x$. Given $\epsilon > 0$ and $N \in \mathbb{N}$, there is K such that $|x_{n_k} - x| < \epsilon$ for every $k \geq K$. Since $n_k \to \infty$, one can choose $k \geq K$ with $n_k > N$; that $m = n_k$ works. □

2.11 Limit Inferior and Limit Superior

Definition 2.7 – Tails of the Sequence

Let (x_n) be a bounded sequence. For each $k \in \mathbb{N}$, the **tail from** k is the set of all terms from k on:

$$A_k = \{x_n : n \geq k\} = \{x_k, x_{k+1}, x_{k+2}, \dots\}.$$

The tails decrease with k: $A_{k+1} \subset A_k$. From them, two auxiliary sequences are defined with the infimum and the supremum of each tail:

- $i_k = \inf(A_k)$: since each tail is a subset of the previous one, its infimum can only go up or stay the same. Thus (i_k) is nondecreasing: $i_k \leq i_{k+1}$.
- $s_k = \sup(A_k)$: analogously, when terms are removed the supremum can only go down. (s_k) is nonincreasing: $s_{k+1} \leq s_k$.

Since (x_n) is bounded, both are monotone and bounded, so that they converge.

Definition 2.8 – Limit Inferior and Superior

The **extreme limits** of (x_n) are defined as the limits of the auxiliary sequences:

(i) Limit inferior:

$$\liminf_{n\to\infty} x_n = \lim_{k\to\infty} i_k = \sup_{k\geq 1} \inf_{n\geq k} x_n.$$

(ii) Limit superior:

$$\limsup_{n\to\infty} x_n = \lim_{k\to\infty} s_k = \inf_{k\geq 1} \sup_{n\geq k} x_n.$$

It always holds that $\liminf x_n \leq \limsup x_n$.

Remark. For unbounded sequences the values $\pm\infty$ are allowed: if (x_n) is not bounded above, $\sup A_k = +\infty$ for every k and $\limsup x_n = +\infty$ (analogously below).

2.12 Characterization by Limit Points

Proposition 2.8 – Characterization

$\limsup x_n$ is the maximum of the limit points of (x_n).

Proof. Let $S = \limsup x_n = \lim_k s_k$ and let P be the set of limit points.

Step 1: $S \in P$. For each k, since $s_k = \sup A_k$, there is some x_n with $n \geq k$ and $s_k - 1/k < x_n \leq s_k$. The choice can be made strictly increasing in n: with $n_1 < \cdots < n_{k-1}$ already chosen, it suffices to require in addition $n \geq n_{k-1} + 1$ and to use the supremum again on the tail $A_{\max(k,n_{k-1}+1)}$ with the same $\eta = 1/k$. Letting $k \to \infty$, since $s_k \to S$ and $|x_{n_k} - s_k| < 1/k$, one gets $x_{n_k} \to S$.

Step 2: if $a \in P$, $a \leq S$. Let $(x_{n_j}) \to a$. For each fixed k, the indices $n_j \geq k$ satisfy $x_{n_j} \leq s_k$. Letting $j \to \infty$ gives $a \leq s_k$, and letting $k \to \infty$, $a \leq S$. □

Analogously, $\liminf x_n$ is the minimum of the limit points.

Theorem 2.5 – Bolzano-Weierstrass

Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Proof. If (x_n) is bounded, $\limsup x_n$ is a real number (not $\pm\infty$) and it is a limit point by Proposition 2.8. By the characterization of limit points (2.4), there is a subsequence that converges to $\limsup x_n$. □

2.13 Cauchy Sequences

Definition 2.9 – Cauchy Sequence

(x_n) is **Cauchy** if for every $\epsilon > 0$ there is n_ϵ such that $|x_n - x_m| < \epsilon$ for all $n, m \geq n_\epsilon$.

Remark. Unlike the limit, no external value a appears: the condition says that the terms accumulate among themselves from a certain n on.

2.14 Boundedness Property

Theorem 2.6 – Cauchy Implies Bounded

Every Cauchy sequence is bounded.

Proof. For $\epsilon = 1$ there is n_1 with $|x_n - x_{n_1}| < 1$ for every $n \geq n_1$, so that $|x_n| \leq |x_{n_1}| + 1$ from n_1 on. The previous ones are finitely many. Taking $M = \max\{|x_1|, \ldots, |x_{n_1-1}|, |x_{n_1}| + 1\}$ yields $|x_n| \leq M$ for every n. □

2.15 Equivalences

Theorem 2.7

For a sequence (x_n) in $\mathbb{R}$ the following are equivalent:

i) (x_n) converges.
ii) (x_n) is Cauchy.
iii) $\liminf x_n = \limsup x_n$.

Proof. The chain $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$ is proved.

(i) ⇒ (ii): If $x_n \to a$, by the triangle inequality $|x_n - x_m| \leq |x_n - a| + |a - x_m| < \epsilon$ for large n, m.

(ii) ⇒ (iii): By contraposition. Suppose $i = \liminf < \limsup = s$ and take $\epsilon = (s-i)/3$. For any k there is $x_m \in A_k$ with $x_m > s - \epsilon$ (because $\sup A_k \geq s$) and $x_n \in A_k$ with $x_n < i + \epsilon$ (because $\inf A_k \leq i$). These two terms satisfy $x_m - x_n > s - i - 2\epsilon = \epsilon$, so that (x_n) is not Cauchy.

(iii) ⇒ (i): Let $a = i = s$. Given $\epsilon > 0$, since $i_k \nearrow a$ and $s_k \searrow a$, there is k_0 with $a - \epsilon < i_{k_0}$ and $s_{k_0} < a + \epsilon$. For $n \geq k_0$,

$$a - \epsilon < i_{k_0} \leq x_n \leq s_{k_0} < a + \epsilon,$$

that is, $x_n \to a$. □

2.16 Solved exercises

Exercise 1

Prove that $\sqrt{2}$ is not rational.

Proof. Argue by contradiction: suppose that $\sqrt{2} \in \mathbb{Q}$, that is, that there are integers p and q (with $q \neq 0$) such that

$$\sqrt{2} = \frac{p}{q}.$$

Without loss of generality, suppose that the fraction $\frac{p}{q}$ is in lowest terms, that is, p and q are coprime. Squaring,

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2,$$

whence p^2 is even. Since the square of an odd number is odd, p must be even, so that $p = 2k$ for some integer k. Substituting,

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2,$$

whence q^2 is even and, by the same reasoning, q is even. Then p and q are both divisible by 2, which contradicts the fraction being in lowest terms. Therefore $\sqrt{2}$ is not rational. □

Exercise 2

Prove that if $x < y + \epsilon$ for every $\epsilon > 0$, then $x \leq y$. Deduce that if $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

Proposition 2.9

If $x < y + \epsilon$ for every $\epsilon > 0$, then $x \leq y$.

Proof. Argue by contradiction: suppose that the conclusion is false, that is, $x > y$. Then $x - y$ is positive and one can take $\epsilon_0 = x - y > 0$. By hypothesis, $x < y + \epsilon$ holds for every $\epsilon > 0$; in particular for ϵ_0:

$$x < y + \epsilon_0 = y + (x - y) = x,$$

a contradiction. Therefore $x \leq y$. □

Corollary 2.1 – Bound for Every ϵ

If $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

Proof. The condition $|x - y| < \epsilon$ is equivalent to $-\epsilon < x - y < \epsilon$, which for every $\epsilon > 0$ gives two inequalities: $x < y + \epsilon$ and $y < x + \epsilon$. Applying Proposition 2.9 to each one yields $x \leq y$ and $y \leq x$, hence $x = y$. □

Exercise 3

Proposition 2.10 – 3.i

Given $x, y \in \mathbb{Q}$ with $x < y$, there is $r \in \mathbb{Q}$ such that $x < r < y$.

Proof. It suffices to take the midpoint between x and y:

$$r = \frac{x+y}{2}.$$

Rationality. Since $x, y \in \mathbb{Q}$, their sum is rational and so is the product by $1/2$, so that $r \in \mathbb{Q}$.

Inequality. From $x < y$ it follows that $2x < x + y$, that is $x < \frac{x+y}{2} = r$; and also $x + y < 2y$, that is $r < y$. Hence $x < r < y$. □

Proposition 2.11 – 3.ii

Given $x, y \in \mathbb{R}$ with $x < y$, there is $r \in \mathbb{Q}$ such that $x < r < y$.

Proof. The idea is to dilate the interval (x, y) by multiplying by a sufficiently large integer n so that the distance between nx and ny exceeds 1, so that there is an integer m between them.

Step 1. Since $y - x > 0$, by the Archimedean Property 2.3 there is $n \in \mathbb{N}$ with

$$\frac{1}{n} < y - x \iff 1 < n(y - x) \iff 1 < ny - nx,$$

that is, the distance between nx and ny is greater than 1.

Step 2. Consider $A = \{k \in \mathbb{Z} : k > nx\}$. By the Archimedean Property it is nonempty and it is bounded below, so that by the Well-Ordering Principle it has a minimum $m = \min(A)$. Being the minimum, $m > nx$ (it is in A) and $m - 1 \leq nx$ (the previous one is not in A).

Step 3. From $m - 1 \leq nx$ one has $m \leq nx + 1$, and using Step 1 ($nx + 1 < ny$),

$$nx < m \leq nx + 1 < ny.$$

Step 4. Dividing by $n > 0$ yields $x < \frac{m}{n} < y$. With $r = \frac{m}{n}$, since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, one has $r \in \mathbb{Q}$. □

Exercise 4

This exercise has already been solved.

Exercise 5

Find the supremum, infimum, maximum and minimum of the following sets and justify (only one is solved; the others are analogous).

(i) Set: $A = (a, b]$ with $a < b$.

- Maximum: b.
- Supremum: b.

- Minimum: Does not exist.
- Infimum: a.

Justification:

- *Supremum/Maximum.* One has $x \leq b$ for every $x \in A$, hence b is an upper bound. Since $b \in (a, b]$, then $b \in A$; being an upper bound that belongs to the set, it is the maximum (and hence the supremum).
- *Infimum.* One has $a < x$ for every $x \in A$, hence a is a lower bound. It is shown that it is the greatest: given $\epsilon > 0$, an $x \in A$ with $x < a + \epsilon$ must be found. Take $x = \min(b, a + \epsilon/2)$, which is strictly greater than a and less than or equal to b, so that it belongs to A, and satisfies $x < a + \epsilon$. By the characterization of the infimum, $a = \inf(A)$.
- *Minimum.* By contradiction, suppose that there is a minimum $m \in A$; then $m > a$. The number $m' = \frac{a+m}{2}$ satisfies $a < m' < m$, so that $m' \in (a, b]$ and it is less than m, which contradicts m being the minimum. Therefore it does not exist.

Exercise 6

If $A \subset \mathbb{R}$ is nonempty and bounded, prove that $\inf A \leq \sup A$. If $A \subset B \subset \mathbb{R}$ and $A \neq \emptyset$, prove the monotonicity properties.

Proposition 2.12

Let $A \subset \mathbb{R}$ be a nonempty bounded set. Then $\inf A \leq \sup A$.

Proof. Since $A \neq \emptyset$, there is $x \in A$. By definition of infimum, $\inf A \leq x$, and by definition of supremum, $x \leq \sup A$. By transitivity, $\inf A \leq x \leq \sup A$. □

Proposition 2.13 – 6.i

If B is bounded above, then so is A, and $\sup A \leq \sup B$.

Proof. *Boundedness.* Let M be an upper bound of B; then $x \leq M$ for every $x \in B$. Since $A \subset B$, every $a \in A$ satisfies $a \leq M$, so that M is an upper bound of A and A is bounded above.

Inequality. Since A is nonempty and bounded, $s_A = \sup A$ exists. Let $s_B = \sup B$; it is an upper bound of B and, since $A \subset B$, also of A. Since s_A is the least upper bound of A,

$$\sup A \leq \sup B.$$

□

Proposition 2.14 – 6.ii

If B is bounded below, then so is A, and $\inf B \leq \inf A$.

Proof. *Boundedness.* Let m be a lower bound of B; then $m \leq x$ for every $x \in B$. Since $A \subset B$, $m \leq a$ for every $a \in A$, so that A is bounded below.

Inequality. Let $i_A = \inf A$ and $i_B = \inf B$. Since i_B is a lower bound of B and $A \subset B$, it is also a lower bound of A; and since i_A is the greatest,

$$\inf B \leq \inf A.$$

□

Proposition 2.15 – 6.iii

If A is not bounded, neither is B.

Proof. It is the contrapositive of the previous properties.

- *Above.* If B were bounded above, by (i) so would be A, against the hypothesis. Hence B is not bounded above.
- *Below.* Analogously, if B had a lower bound, A would inherit it (by (ii)), a contradiction.

□

Exercise 7

Prove properties of the supremum and infimum under reflection and scaling.

Proposition 2.16 – 7.i

If A is bounded above, then $-A = \{-x : x \in A\}$ is bounded below and

$$\inf(-A) = -\sup A$$

Proof. Let $s = \sup A$. It must be proved that $-s$ is a lower bound of $-A$ and that it is the greatest.

It is a lower bound. Since s is the supremum of A, $x \leq s$ for every $x \in A$; multiplying by -1, $-x \geq -s$ for every $x \in A$. That is, every $y = -x \in -A$ satisfies $y \geq -s$, so that $-s$ is a lower bound of $-A$.

It is the greatest lower bound. Let t be another lower bound of $-A$, that is, $y \geq t$ for every $y \in -A$. With $y = -x$, $-x \geq t$ for every $x \in A$, that is $x \leq -t$, so that $-t$ is an upper bound of A. Since s is the least upper bound, $s \leq -t$, and multiplying by -1, $-s \geq t$. Hence $-s = \inf(-A)$. □

Proposition 2.17 – 7.ii

If $c > 0$ and A is bounded above, then $cA = \{cx : x \in A\}$ is bounded above and

$$\sup(cA) = c \cdot \sup A$$

Proof. Let $s = \sup A$.

It is an upper bound. One has $x \leq s$ for every $x \in A$; since $c > 0$, $cx \leq cs$ for every $x \in A$, so that cs is an upper bound of cA.

It is the least upper bound. Let M be an upper bound of cA, that is, $cx \leq M$ for every $x \in A$. Since $c > 0$, $x \leq M/c$ for every $x \in A$, so that M/c is an upper bound of A; since s is the least, $s \leq M/c$, and multiplying by c, $cs \leq M$. Hence $cs = \sup(cA)$. □

Exercise 8

This exercise has already been solved.

Exercise 9

Prove that a sequence (x_n) converges to a if and only if every subsequence of (x_n) converges to a.

Proof. Both implications are proved.

($\Rightarrow$). Suppose that $x_n \to a$ and let (x_{n_k}) be an arbitrary subsequence. Given $\epsilon > 0$, there is N such that $|x_n - a| < \epsilon$ for $n \geq N$. Since the indices of a subsequence satisfy $n_k \geq k$, for $k \geq N$ one has $n_k \geq N$ and therefore $|x_{n_k} - a| < \epsilon$. Hence $x_{n_k} \to a$.

($\Leftarrow$). The original sequence is a subsequence of itself (with $n_k = k$), so that by hypothesis it converges to a.

An alternative proof of the converse, by contraposition: suppose that x_n does not converge to a. Then there is $\epsilon > 0$ such that for every N there is $n > N$ with $|x_n - a| \geq \epsilon$. A subsequence is constructed by choosing those terms: n_1 with $|x_{n_1} - a| \geq \epsilon$, then $n_2 > n_1$ with $|x_{n_2} - a| \geq \epsilon$, and so on. All its terms are at distance $\geq \epsilon$ from a, so that (x_{n_k}) does not converge to a, which contradicts the hypothesis. Therefore $x_n \to a$. □

Exercise 10

Prove that if (x_n) converges to a, then a is the only limit point of the sequence (x_n).

Proof. It is already known that if $x_n \to a$ then a is a limit point; it remains to prove uniqueness. Argue by contradiction: suppose that there is another limit point $b \neq a$.

- By the characterization of limit points (Theorem 2.4), there is a subsequence $(x_{n_k}) \to b$.
- Since $x_n \to a$, every subsequence converges to a; in particular $(x_{n_k}) \to a$.
- By uniqueness of the limit (Proposition 2.6), from $x_{n_k} \to b$ and $x_{n_k} \to a$ it follows that $b = a$, against the assumption.

Hence a is the only limit point. □

Exercise 11

Prove that there is a sequence (x_n) such that every $a \in \mathbb{R}$ is a limit point of (x_n).

Proof. Since $\mathbb{Q}$ is countable, there is a bijection $f : \mathbb{N} \to \mathbb{Q}$; define (x_n) as an enumeration of the rationals,

$$(x_n)_{n\in\mathbb{N}} = (q_1, q_2, q_3, \dots), \qquad \{q_n : n \in \mathbb{N}\} = \mathbb{Q}.$$

Let $a \in \mathbb{R}$, $\epsilon > 0$ and $k \in \mathbb{N}$; an $n > k$ with $|x_n - a| < \epsilon$ must be found. By the density of $\mathbb{Q}$ in $\mathbb{R}$ (2.4), the set $S = \{q \in \mathbb{Q} : |q - a| < \epsilon\}$ is infinite, and since (x_n) runs through all the rationals, there are infinitely many indices n with $x_n \in S$; in particular one with $n > k$. For that n, $|x_n - a| < \epsilon$. Thus a is a limit point, and since a was arbitrary, every real is one. □

Exercise 12

This exercise has already been solved.

Exercise 13

Prove that if (x_n) is Cauchy, and a is a limit point of (x_n), then (x_n) converges to a.

Proof. It must be proved that $x_n \to a$. Let $\epsilon > 0$.

Cauchy condition. Since (x_n) is Cauchy, there is n_0 such that $|x_n - x_m| < \frac{\epsilon}{2}$ for all $n, m \geq n_0$.

Limit point condition. Since a is a limit point, there is a subsequence $(x_{n_k}) \to a$, so that there is K with $|x_{n_k} - a| < \frac{\epsilon}{2}$ for $k \geq K$.

Combination. Since (n_k) is strictly increasing, it eventually exceeds n_0; take k_ϵ with $k_\epsilon \geq n_0$ and $k_\epsilon \geq K$. For every $n \geq n_0$, by the triangle inequality,

$$|x_n - a| \leq |x_n - x_{k_\epsilon}| + |x_{k_\epsilon} - a|.$$

The first term is $< \frac{\epsilon}{2}$ (because n and n_{k_ϵ} are $\geq n_0$ and the sequence is Cauchy) and so is the second (because $n_{k_\epsilon} \geq K$). Adding,

$$|x_n - a| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore $x_n \to a$. □

3 Metric Spaces

3.1 Definition and Properties

Definition 3.1 – Metric

A **metric** on X is a function $d : X \times X \to \mathbb{R}$ that satisfies, for all $x, y, z \in X$:

(i) Positivity: $d(x, y) \geq 0$.
(ii) Nondegeneracy: $d(x, y) = 0 \iff x = y$.
(iii) Symmetry: $d(x, y) = d(y, x)$.
(iv) Triangle inequality: $d(x, y) \leq d(x, z) + d(z, y)$.

A **metric space** is a pair (X, d) with X nonempty.

3.2 Examples of Metric Spaces

Example 3.1 – Metrics on $\mathbb{R}^n$

On $\mathbb{R}^n$ there are several distances: the Euclidean $d_2(x, y) = \sqrt{\sum (x_i - y_i)^2}$, the sum distance $d_1(x, y) = \sum |x_i - y_i|$ and the maximum distance $d_\infty(x, y) = \max_i |x_i - y_i|$. They come from the norms $\| \cdot \|_2$, $\| \cdot \|_1$ and $\| \cdot \|_\infty$.

Example 3.2 – Discrete Metric

$d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ otherwise. All points are at distance 1 from each other.

Example 3.3 – Post Office Metric

On $\mathbb{R}^2$, every path goes through the origin unless the points coincide:

$$d(x, y) = \begin{cases} 0 & x = y, \\ \|x\| + \|y\| & x \neq y. \end{cases}$$

Example 3.4 – Function Space $C[0, 1]$

$C[0, 1]$ with the supremum distance $d(f, g) = \max_{t \in [0,1]} |f(t) - g(t)|$. It measures uniform convergence.

3.3 Open Balls

Definition 3.2 – Open Ball

The **open ball** with center a and radius $r > 0$ is $B(a, r) = \{x \in X : d(a, x) < r\}$.

Note that every ball contains its center ($d(a, a) = 0 < r$), so that it is never empty.

3.4 Open Sets

Definition 3.3 – Open Set

$A \subset X$ is **open** if for every $x \in A$ there is $r > 0$ with $B(x, r) \subset A$.

Example 3.5 – Discrete Metric

Every subset A of a discrete space is open: the ball $B(x, 1/2) = \{x\}$ (because the distances are 0 or 1), and $\{x\} \subset A$.

Example 3.6 – Function Space $C[0, 1]$

In $C[0, 1]$ with the sup norm, $A = \{f : f(1/2) \neq 3\}$ is open.

Proof. Given $f \in A$, take $r = |f(1/2) - 3| > 0$. If $g \in B(f, r)$, then $|f(1/2) - g(1/2)| \leq \|f - g\|_\infty < r = |f(1/2) - 3|$, so that $g(1/2) \neq 3$ and $g \in A$. □

3.5 Properties of Open Sets

Proposition 3.1 – Open Balls are Open

Every open ball $B(a, r)$ is an open set.

Proof. Let $x \in B(a, r)$, so that $d(a, x) < r$. Take $r_x = r - d(a, x) > 0$ (the distance to the edge); it is shown that $B(x, r_x) \subset B(a, r)$. If $y \in B(x, r_x)$, by the triangle inequality

$$d(a, y) \leq d(a, x) + d(x, y) < d(a, x) + r_x = r,$$

that is, $y \in B(a, r)$. □

Proposition 3.2 – Properties of the Topology

In every metric space:

(i) $\emptyset$ and X are open.
(ii) An arbitrary union of open sets is open.
(iii) A finite intersection of open sets is open.

Proof. $\emptyset$ is open vacuously and X by definition (any ball stays in X).

Union: Let $U = \bigcup_i U_i$ and $x \in U$. There is k with $x \in U_k$; since U_k is open, there is $r > 0$ with $B(x, r) \subset U_k \subset U$.

Finite intersection: Let $W = \bigcap_{j=1}^{n} U_j$ and $x \in W$. For each j there is $r_j > 0$ with $B(x, r_j) \subset U_j$. Taking $r = \min_j r_j > 0$ (it is positive because there are finitely many), $B(x, r) \subset U_j$ for every j, so that $B(x, r) \subset W$. □

Remark. An infinite intersection of open sets may fail to be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

3.6 Topological Spaces

The structure of a metric space depends on the distance. Retaining only the previous properties and forgetting the distance yields a more general structure.

Definition 3.4 – Topology

A **topology** on X is a family τ of subsets (the **open sets**) such that:
(i) $\emptyset, X \in \tau$.
(ii) An arbitrary union of elements of τ is in τ.
(iii) A finite intersection of elements of τ is in τ.
(X, τ) is called a **topological space**.

3.7 Relation to Metric Spaces

Every metric space (X, d) induces a topology τ_d (the open sets according to the metric), so that every metric space is topological. The converse is false: there are topologies that do not come from any metric.

3.8 Closed Sets

Definition 3.5 – Closed Set

$F \subset X$ is **closed** if F^c is open.

Proposition 3.3 – Properties of Closed Sets

By De Morgan, closed sets are dual to open sets:
(i) $\emptyset$ and X are closed.
(ii) An arbitrary intersection of closed sets is closed.
(iii) A finite union of closed sets is closed.

3.9 Interior and Closure

Definition 3.6 – Interior Point and Interior

$x \in A$ is an **interior point** if there is $r > 0$ with $B(x, r) \subset A$. The **interior** A° is the set of all such points.

Definition 3.7 – Closure Point and Closure

$x \in X$ is a **closure point** of A if every ball $B(x, r)$ meets A. The **closure** $\overline{A}$ is the set of all such points.

3.10 Inclusion Properties

Proposition 3.4 – Basic Inclusions

$A^\circ \subseteq A \subseteq \overline{A}$.

Proof. If $x \in A^\circ$, there is a ball around x inside A, in particular $x \in A$. If $x \in A$, the ball $B(x,r)$ always contains $x \in A$, so that x is a closure point. □

3.11 Duality between Interior and Closure

Interior and closure are dual under complement: what lies outside the interior is the closure of the exterior.

Proposition 3.5 – Duality by Complement

For every $A \subset X$:

$$(A^\circ)^c = \overline{A^c}, \qquad (\overline{A})^c = (A^c)^\circ.$$

Proof. For the first, $x \in (A^\circ)^c$ if and only if no ball around x is contained in A, if and only if every ball has a point of A^c, if and only if $x \in \overline{A^c}$.

The second is the first applied to A^c. □

3.12 Extremal Properties

Proposition 3.6 – The Interior is the Largest Open Set

A° is open, it is contained in A, and every open $U \subseteq A$ satisfies $U \subseteq A^\circ$.

Proof. *Open:* If $x \in A^\circ$ with $B(x,r) \subset A$, for any $y \in B(x,r)$ there is $\delta > 0$ with $B(y,\delta) \subset B(x,r) \subset A$ (because balls are open), so that $y \in A^\circ$. Hence $B(x,r) \subset A^\circ$.

Maximality: If U is open and $U \subseteq A$, given $x \in U$ there is $r > 0$ with $B(x,r) \subset U \subseteq A$, so that $x \in A^\circ$. □

Proposition 3.7 – The Closure is the Smallest Closed Set

$\overline{A}$ is closed, it contains A, and every closed $F \supseteq A$ satisfies $F \supseteq \overline{A}$.

Proof. *Closed:* $(\overline{A})^c = (A^c)^\circ$ is open, so that $\overline{A}$ is closed.

Minimality: If F is closed with $A \subseteq F$, then $F^c \subseteq A^c$ is open, so that $F^c \subseteq (A^c)^\circ = (\overline{A})^c$, which gives $\overline{A} \subseteq F$. □

3.13 Characterization Corollaries

Corollary 3.1 – Characterization of Open Sets

A is open if and only if $A = A^\circ$.

Proof. Both implications are proved.
(⇒) $A^\circ \subseteq A$ always, and since A is an open set contained in itself, by maximality $A \subseteq A^\circ$.
(⇐) A° is always open. □

Corollary 3.2 – Characterization of Closed Sets

A is closed if and only if $A = \overline{A}$.

Proof. Analogous: by minimality, every closed set that contains itself equals its closure, and $\overline{A}$ is always closed. □

Corollary 3.3 – Closed Sets via Sequences

$F \subset X$ is closed if and only if every convergent sequence $(x_n) \subset F$ has its limit in F.

Proof. If F is closed and $x_n \to x$ with $x_n \in F$, $x \in \overline{F} = F$. Conversely, given $x \in \overline{F}$ there is $(x_n) \subset F$ with $x_n \to x$ (characterization by sequences, 3.8), and by hypothesis $x \in F$, so that $\overline{F} \subseteq F$ and F is closed. □

3.14 Accumulation Point and Boundary

Definition 3.8 – Accumulation Point

$x \in X$ is an **accumulation point** of A if every ball around x has some point of A *different* from x:

$$\forall r > 0, \quad (B(x, r) \cap A) \setminus \{x\} \neq \emptyset.$$

Remark. Every accumulation point is in the closure, but not conversely: the **isolated points** (points $x \in A$ such that some ball $B(x, r)$ meets A only in $\{x\}$) live in $\overline{A}$ but are not accumulation points. Calling A' the set of accumulation points (the **derived set**), one has $\overline{A} = A \cup A'$.

Definition 3.9 – Boundary Point and Boundary

$x \in X$ is a **boundary point** of A if every ball centered at x touches both A and A^c. The **boundary** is $\partial A = \{x : x \text{ is a boundary point}\}$. Equivalently, $\partial A = \overline{A} \cap \overline{A^c} = \overline{A} \setminus A^\circ$.

3.15 Sequences and Convergence

Definition 3.10 – Sequence

A **sequence** in (X, d) is a function $x : \mathbb{N} \to X$, denoted $(x_n)_{n \in \mathbb{N}}$.

Definition 3.11 – Convergence

(x_n) **converges** to $x \in X$ (written $x_n \to x$) if for every $\epsilon > 0$ there is n_ϵ such that $d(x_n, x) < \epsilon$ for every $n \geq n_\epsilon$. In terms of balls: from a certain n_ϵ on, all the terms live in $B(x, \epsilon)$.

Definition 3.12 – Subsequence

A **subsequence** of (x_n) is $(x_{n_k})_{k \in \mathbb{N}}$ where $k \mapsto n_k$ is strictly increasing.

3.16 Cauchy Sequences and Completeness

Definition 3.13 – Cauchy Sequence

(x_n) is **Cauchy** if for every $\epsilon > 0$ there is n_ϵ such that $d(x_n, x_m) < \epsilon$ for all $n, m \geq n_\epsilon$.

Definition 3.14 – Complete Space

A metric space (X, d) is called **complete** if every Cauchy sequence in X converges to a point of X.

Example 3.7 – Examples of Completeness

- The real numbers $\mathbb{R}$ with the usual distance are a complete space (Completeness Axiom).
- The rational numbers $\mathbb{Q}$ are not complete.
- A space with the discrete metric is complete (the only Cauchy sequences are the ones that become constant from some point on).

3.17 Characterization of the Closure by Sequences

Proposition 3.8 – Closure and Sequences

$x \in \overline{A}$ if and only if there is $(a_n) \subset A$ with $a_n \to x$.

Proof. Both implications are proved.

($\Rightarrow$) If $x \in \overline{A}$, for each n the ball $B(x, 1/n)$ meets A; choose a_n in that intersection. $d(a_n, x) < 1/n \to 0$.

($\Leftarrow$) If $a_n \to x$ with $a_n \in A$, given $r > 0$ some $a_{n_0} \in B(x, r) \cap A$, so that the intersection is not empty. □

3.18 Characterization of Accumulation Points

Proposition 3.9 – Accumulation and Sequences

x is an accumulation point of A if and only if there is $(a_n) \subset A$ of distinct points with $a_n \to x$.

Proof. Both implications are proved.

($\Rightarrow$) The a_n are constructed with decreasing radius: take $a_1 \in B(x, 1) \cap A \setminus \{x\}$ and, given $a_1, \ldots, a_n$, define $r_{n+1} = d(x, a_n)/2 > 0$ and choose $a_{n+1} \in B(x, r_{n+1}) \cap A \setminus \{x\}$. By construction $d(x, a_{n+1}) < d(x, a_n)/2$, so that the distances go down to zero (the a_n are distinct from each other because $d(x, a_{n+1}) < d(x, a_n)$).

($\Leftarrow$) Given $r > 0$, there is some $a_{n_0} \in B(x, r) \cap A$ with $a_{n_0} \neq x$ (if $a_{n_0} = x$, take a_{n_0+1}). □

3.19 Completeness and Closed Sets

Proposition 3.10 – Closed in Complete is Complete

If (X, d) is complete and $F \subset X$ is closed, (F, d) is complete.

Proof. A Cauchy sequence in F is Cauchy in X, and since X is complete it converges to $x \in X$. By the characterization of the closure by sequences (3.8), $x \in \overline{F} = F$. □

3.20 Diameter of a Set

Definition 3.15 – Diameter

For $A \subset X$ nonempty and bounded, the diameter is

$$\operatorname{diam}(A) = \sup\{d(x, y) : x, y \in A\}.$$

Remark. In $\mathbb{R}^n$ with the Euclidean metric the diameter of a ball of radius r is $2r$, but in general $0 \leq \operatorname{diam}(B(a, r)) \leq 2r$, and the inequality may be strict (for example in the discrete metric, $\operatorname{diam}(B(a, 1)) = \operatorname{diam}(\{a\}) = 0$).

Remark. One has $\operatorname{diam}(A) = \operatorname{diam}(\overline{A})$: if $x', y' \in \overline{A}$, take sequences $x_n, y_n \in A$ with $x_n \to x'$, $y_n \to y'$, and $d(x', y') = \lim d(x_n, y_n) \leq \operatorname{diam}(A)$, so that $\operatorname{diam}(\overline{A}) \leq \operatorname{diam}(A)$; the other inequality follows from inclusion.

3.21 Density and Separability

Definition 3.16 – Dense Set

$A \subset X$ is dense if $\overline{A} = X$: any point of the space can be approximated by points of A.

Definition 3.17 – Separable Space

X is separable if it has a countable dense subset. For example, $\mathbb{R}$ is separable because $\mathbb{Q}$ is dense and countable.

3.22 Characterization of Dense Sets

Proposition 3.11 – Characterization by Open Sets

A is dense if and only if it meets every nonempty open set.

Proof. Both implications are proved.

(⇒) Let U be a nonempty open set and $x \in U$, with $B(x, r) \subset U$. Since $x \in \overline{A}$, $B(x, r) \cap A \neq \emptyset$, and then $U \cap A \neq \emptyset$.

(⇐) Given $x \in X$ and $r > 0$, the ball $B(x, r)$ is open and nonempty, so that it meets A. That says $x \in \overline{A}$. □

3.23 Cantor's Intersection Theorem

Theorem 3.1 – Cantor's Theorem

Let (X, d) be a complete metric space. Let (A_n) be a sequence of subsets of X satisfying the following conditions:

(i) They are closed: $A_n = \overline{A_n}$ for every n.
(ii) They are nonempty: $A_n \neq \emptyset$.
(iii) They are decreasing: $A_{n+1} \subseteq A_n$.
(iv) Their diameters tend to zero: $\operatorname{diam}(A_n) \to 0$ as $n \to \infty$.

Then the intersection of all the sets contains exactly one point:

$$\bigcap_{n=1}^{\infty} A_n = \{x\}$$

Proof. Choose $a_n \in A_n$ for each n. If $n, m \geq N$, since the family is decreasing $a_n, a_m \in A_N$, so that $d(a_n, a_m) \leq \operatorname{diam}(A_N)$. Since $\operatorname{diam}(A_N) \to 0$, (a_n) is Cauchy and by completeness it converges to some $x \in X$.

For each fixed k, the tail $(a_n)_{n \geq k}$ lives in A_k and converges to x, so that $x \in \overline{A_k} = A_k$. Hence $x \in \bigcap A_n$.

Uniqueness: if x, y are both in the intersection, $d(x, y) \leq \operatorname{diam}(A_n) \to 0$, so that $x = y$. □

3.24 Why the diameter must tend to zero

Example 3.8 – Empty Intersection in a Complete Space

Consider the space $X = \mathbb{N}$ with the metric

$$d(n,m) = \begin{cases} 0 & \text{if } n = m \\ 1 + \frac{1}{n} + \frac{1}{m} & \text{if } n \neq m \end{cases}$$

This space is complete (the only Cauchy sequences are the ones that are eventually constant).
Define the sequence of closed sets:

$$A_n = \{k \in \mathbb{N} : k \geq n\} = \{n, n+1, n+2, \dots\}$$

These sets satisfy:

- They are closed.
- They are nonempty.
- They are decreasing ($A_{n+1} \subset A_n$).

However, their intersection is empty:

$$\bigcap_{n=1}^{\infty} A_n = \emptyset$$

The theorem fails because the diameters do not tend to zero. The diameter of A_n is the supremum of the distances between $k, j \geq n$. When k, j are very large, the distance approaches 1.

$$\operatorname{diam}(A_n) \to 1 \neq 0$$

3.25 Relation between Balls and Closure

Proposition 3.12 – Inclusion of Balls

If $0 < r' < r$, then $\overline{B(x,r')} \subseteq B(x,r)$.

Proof. Let $y \in \overline{B(x,r')}$ and $\epsilon = r - r' > 0$. Since y is a closure point, $B(y,\epsilon) \cap B(x,r') \neq \emptyset$; take z there. By the triangle inequality,

$$d(x,y) \leq d(x,z) + d(z,y) < r' + \epsilon = r,$$

so that $y \in B(x,r)$. □

3.26 Closed Ball vs. Closure of the Ball

Example 3.9 – Counterexample in the Discrete Metric

In a discrete space X with more than one point, looking at radius $r = 1$:

- $B(x, 1) = \{x\}$ (only $d(x, x) = 0 < 1$).
- $\overline{B(x, 1)} = \overline{\{x\}} = \{x\}$ (singletons are closed).
- Closed ball $\{y : d(x, y) \leq 1\} = X$ (all distances are 0 or 1).

Thus, $\overline{B(x, 1)} = \{x\} \neq X = \{y : d(x, y) \leq 1\}$: the closure of the ball is much smaller than the closed ball.

3.27 Baire Category Theorem

Theorem 3.2 – Baire's Theorem (Version 1)

If X is a complete metric space and (U_n) a countable family of dense open sets, then $G = \bigcap_n U_n$ is dense in X.

Proof. It suffices to see that G meets every nonempty open set. Let V be a nonempty open set. Nested balls are constructed: since U_1 is dense, $V \cap U_1$ is a nonempty open set, so that one chooses B_1 with $\overline{B_1} \subset V \cap U_1$ and radius < 1. Inductively, with B_{n-1} already chosen, take B_n with $\overline{B_n} \subset B_{n-1} \cap U_n$ and radius $< 1/n$.

The $\overline{B_n}$ are closed, nonempty, decreasing and with diameter $\to 0$. By Cantor's Theorem 3.1, $\bigcap_n \overline{B_n} = \{x\}$. And $x \in \overline{B_1} \subset V$ and $x \in \overline{B_n} \subset U_n$ for every n, so that $x \in V \cap G$. □

3.28 Version 2: Union of Closed Sets

Definition 3.18 – Categories

- A is **nowhere dense** if $(\overline{A})^\circ = \emptyset$.
- A is of the **first category** if it is a countable union of nowhere dense sets.
- A is of the **second category** if it is not of the first.

Theorem 3.3 – Baire's Theorem (Version 2)

Every complete metric space is of the second category. That is, if X is complete and $X = \bigcup_n F_n$ with each F_n closed, at least one has nonempty interior.

Proof. Argue by contradiction: suppose F_n closed with $F_n^\circ = \emptyset$ and $\bigcup F_n = X$. Then $U_n = F_n^c$ is open and, since F_n has empty interior, U_n is dense. By version 1, $\bigcap U_n$ is dense, in particular nonempty. But by De Morgan $\bigcap U_n = (\bigcup F_n)^c = X^c = \emptyset$, a contradiction. □

3.29 Application: Nowhere Differentiable Continuous Functions

An application of Baire: the continuous functions that are differentiable *at no point* are dense in $C[0, 1]$. Intuition suggests picturing them as smooth curves, but topologically the rule is the ones that have no derivative at any point.

Work in $X = C[0,1]$ with the supremum norm, which is complete. For each $r \in \mathbb{Q} \cap [0,1]$ and $n, m \in \mathbb{N}$ define the set of functions with large slope near r:

$$U_{r,n,m} = \left\{ f \in C[0,1] : \sup_{t \in (0,1/n)} \left| \frac{f(r+t) - f(r)}{t} \right| > m \right\}.$$

It will be shown that each $U_{r,n,m}$ is open and dense, and then Baire is applied.

1. They are open

Let $f \in U_{r,n,m}$. By definition of the supremum there is $t_0 \in (0, 1/n)$ with

$$K = \left| \frac{f(r+t_0) - f(r)}{t_0} \right| > m.$$

Take $g \in B(f, \epsilon)$ (with ϵ small). Its difference quotient at the same t_0 satisfies

$$\left| \frac{g(r+t_0) - g(r)}{t_0} \right| \geq K - \frac{2\epsilon}{t_0},$$

which is still $> m$ if ϵ is sufficiently small. Thus $B(f, \epsilon) \subset U_{r,n,m}$.

2. They are dense

It must be shown that for every $h \in C[0,1]$ and every $\epsilon > 0$, $B(h, \epsilon) \cap U_{r,n,m} \neq \emptyset$. The idea is to perturb h on a small interval $[r, r+t]$, forcing it to have slope greater than m there.

Define $g(r) = h(r)$ and $g(r+t) = h(r) + (m+1)t$. The difference quotient becomes $(g(r+t) - g(r))/t = m + 1 > m$, which is what is sought.

It remains to have $\|g - h\|_\infty < \epsilon$. At the worst point,

$$|g(r+t) - h(r+t)| \leq |h(r) - h(r+t)| + (m+1)t.$$

Since h is continuous at r, both terms go to zero as $t \to 0$. Choose t_0 with $|h(r) - h(r+t_0)| < \epsilon/2$ and $(m+1)t_0 < \epsilon/2$. Define g equal to h outside $[r, r+t_0]$ and linearly connecting $(r, h(r))$ and $(r+t_0, g(r+t_0))$ inside; by continuity of h and because t_0 is small, the linear perturbation stays inside the tube of radius ϵ.

Conclusion

The family $\{U_{r,n,m}\}$ with $r \in \mathbb{Q} \cap [0,1]$ and $n, m \in \mathbb{N}$ is a countable family of dense open sets, so that its intersection $G = \bigcap_{r,n,m} U_{r,n,m}$ is dense in $C[0,1]$.

If $f \in G$, it is not differentiable at any $x \in (0,1)$: if it were, the difference quotient would be bounded on a neighborhood of x; but by density of $\mathbb{Q}$ and because f lies in each $U_{r,n,m}$, arbitrarily large difference quotients appear near x. Contradiction.

3.30 Pathological examples

Geometric intuition sometimes fails. A function is constructed that is strictly increasing, discontinuous at *every* rational and continuous at every irrational.

Construction. Since $\mathbb{Q}$ is countable, fix an enumeration $\{r_1, r_2, \dots\}$. For $x \in \mathbb{R}$ let $S_x = \{k : r_k < x\}$ and

$$f(x) = \sum_{k \in S_x} \frac{1}{2^k}.$$

It is a subseries of the geometric series, so that it converges and $f(x) \in [0, 1]$.

Strictly increasing. If $x < y$, $S_x \subseteq S_y$, and by density of $\mathbb{Q}$ (2.4) there is r_m with $x < r_m < y$, so that $m \in S_y \setminus S_x$ and $f(y) \geq f(x) + 1/2^m$.

Discontinuous at the rationals. Take r_m. By definition $m \notin S_{r_m}$ (because $r_m \not< r_m$). But for every $t > r_m$ one has $m \in S_t$, so that

$$f(t) \geq f(r_m) + \frac{1}{2^m},$$

and the right-hand limit exceeds $f(r_m)$ by at least $1/2^m$.

Continuous at the irrationals. Let $a \notin \mathbb{Q}$. Given $\epsilon > 0$ take N with $\sum_{k>N} 1/2^k < \epsilon$. Since a is irrational, it is at positive distance from the first N rationals; let $\delta = \min_{k \leq N} |a - r_k| > 0$. None of $r_1, \dots, r_N$ appears in $(a - \delta, a + \delta)$, so that for x in that neighborhood, S_x and S_a coincide on the indices $\leq N$. The difference $|f(x) - f(a)|$ uses only $k > N$ and is bounded by $\sum_{k>N} 1/2^k < \epsilon$.

3.31 The Cantor ternary set

It is a subset of $[0, 1]$ that is closed, with empty interior and length zero, but with as many points as $\mathbb{R}$.

Construction. Recursively, always removing the middle third:

- $F_0 = [0, 1]$.
- $F_1 = [0, 1/3] \cup [2/3, 1]$ ($(1/3, 2/3)$ is removed).
- $F_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$.
- At step n, a union of 2^n closed intervals of length $1/3^n$.

$C = \bigcap_n F_n$.

Closed. Each F_n is a finite union of closed sets, so that F_n is closed; C, an intersection of closed sets, is closed as well.

Measure zero (empty interior). At step k, 2^{k-1} intervals of length $1/3^k$ are removed, so that the total length removed is

$$\sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k} = \frac{1}{3} \sum_{k=0}^{\infty} (2/3)^k = 1.$$

A total length of 1 was removed from an interval of length 1: C contains no interval.

Uncountable. The ternary expansion is used. If $x = 0.a_1 a_2 \ldots_{(3)}$ with $a_k \in \{0, 1, 2\}$, at step n the x whose $a_n = 1$ are removed (the middle third at each level). Thus, the $x \in C$ are the ones that have an expansion with $a_k \in \{0, 2\}$ only. The bijection $a_k \mapsto a_k/2$ identifies C with $[0, 1]$ in binary, so that $|C| = c$.

3.32 Existence of Utility Functions

In consumer theory, the points of $\mathbb{R}^n$ are bundles of goods and the preference relation $\succeq$ orders the bundles. Three axioms are required of it:

- *Transitivity:* $x \succeq y$ and $y \succeq z$ imply $x \succeq z$.
- *Rationality:* for all x, y, $x \succeq y$ or $y \succeq x$.
- *Continuity:* if $x \succ y$, the set $\{w : x \succ w \succ y\}$ is a nonempty open set.

Continuity prevents abrupt changes: what is near x is preferred to what is near y.

Theorem 3.4 – Trout-Rader Theorem

If $\succeq$ satisfies transitivity, rationality and continuity, there is $u : \mathbb{R}^n \to \mathbb{R}$ with $x \succeq y \iff u(x) \geq u(y)$.

Proof. Fix an enumeration $\{r_1, r_2, \dots\}$ of $\mathbb{Q}^n$ (countable dense in $\mathbb{R}^n$). Define $S_x = \{k : x \succeq r_k\}$ and

$$u(x) = \sum_{k \in S_x} \frac{1}{2^k}.$$

It is a subseries of the geometric series, so that it converges.

($\Rightarrow$) If $x \succeq y$ and $k \in S_y$, by transitivity $x \succeq r_k$, so that $k \in S_x$. Hence $S_y \subseteq S_x$ and $u(y) \leq u(x)$.

($\Leftarrow$) By strict contraposition: if $y \succ x$, it must be shown that $u(y) > u(x)$. By continuity $\{w : y \succ w \succ x\}$ is a nonempty open set; by density of $\mathbb{Q}^n$ there is some r_m in it. Then $m \in S_y$ and $m \notin S_x$ (because $r_m \succ x$ negates $x \succeq r_m$). Since $S_x \subseteq S_y$ (by the previous implication), $u(y) \geq u(x) + 1/2^m > u(x)$. □

Remark. The u obtained is neither continuous nor differentiable; it is only a numerical label that respects the order. To work analytically (differentiate, optimize) more must be required.

3.33 Solved Exercises

Exercise 1

Prove that the following pairs are metric spaces and draw an open ball in each case.

i.

$$X = \mathbb{R}, \quad d(x, y) = |x - y|$$

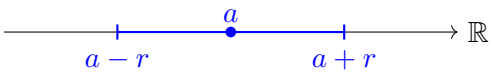
Proof. (i) The absolute value is nonnegative and vanishes only if the argument is zero ($x - y = 0 \implies x = y$).

(ii) $|x - y| = |-(y - x)| = |y - x|$.

(iii) $|x - z| = |x - y + y - z| \leq |x - y| + |y - z|$ (triangle inequality in $\mathbb{R}$).

□

Open Ball: $B(a, r) = \{x \in \mathbb{R} : |x - a| < r\} = (a - r, a + r)$. It is an open interval.



ii.

$$X = \mathbb{R}^n, \quad d_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Proof. (i) The square root of a sum of squares is nonnegative and vanishes only if each term is zero ($x_i = y_i$).

(ii) $(x_i - y_i)^2 = (y_i - x_i)^2$, so that the sum is the same.

(iii) The triangle inequality is Minkowski's inequality for sums: $\|x - z\|_2 \leq \|x - y\|_2 + \|y - z\|_2$.

□

Open Ball ($n = 2$): It is the interior of an ordinary circle.

iii.

$$X = \mathbb{R}^n, \quad d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

Proof. Properties 1 and 2 follow from those of the absolute value in each coordinate. For the triangle inequality, in each component i one has $|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$; summing over i yields the inequality for the metric. □

Open Ball ($n = 2$): $|x_1 - a_1| + |x_2 - a_2| < r$. It is a square rotated by 45 degrees.

iv.

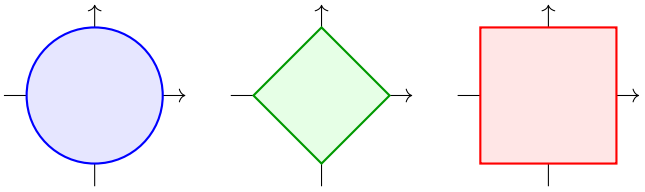
$$X = \mathbb{R}^n, \quad d_\infty(x, y) = \max_{1 \le i \le n} |x_i - y_i|$$

Proof. The maximum of nonnegative values is nonnegative and vanishes only if all of them do, and symmetry is inherited from the absolute value. For the triangle inequality, let k be the index where the maximum of $|x_i - z_i|$ is attained; then

$$d_\infty(x, z) = |x_k - z_k| \le |x_k - y_k| + |y_k - z_k| \le d_\infty(x, y) + d_\infty(y, z),$$

since each term is less than or equal to its respective maximum. □

Open Ball ($n = 2$): $\max(|x_1 - a_1|, |x_2 - a_2|) < r$. It is a square with sides parallel to the axes.



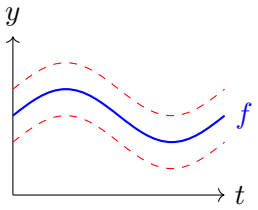
ii. Euclidean iii. Sum iv. Maximum

v.

$$X = C([0, 1]), \quad d(f, g) = \max_{t \in [0,1]} |f(t) - g(t)|$$

Proof. Analogous to the maximum metric on $\mathbb{R}^n$, taking the maximum over the continuous interval $t \in [0, 1]$. □

Open Ball: It is a strip or tube of width $2r$ centered at the function f.



vi.

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

Proof. Properties 1 and 2 hold by definition. For the triangle inequality, $d(x, z) \le d(x, y) + d(y, z)$: if $x \neq z$ then $d(x, z) = 1$, and for the sum on the right to be 0 one would need $x = y$ and $y = z$, which gives $x = z$, a contradiction; hence the right-hand side adds up to at least 1. □

Open Ball:

- If $r \le 1$: $B(x, r) = \{x\}$.
- If $r > 1$: $B(x, r) = X$.

Exercise 2

Given fixed $x \in X$ and $r > 0$, prove that the following sets are open.

i. The set $\{y \in X : d(x, y) < r\}$

This exercise has already been solved.

ii. The set $A = \{y \in X : d(x, y) > r\}$

Proof. Let $y_0 \in A$, that is, $d(x, y_0) = d > r$. A $\delta > 0$ such that $B(y_0, \delta) \subset A$ must be found. Define

$$\delta = d(x, y_0) - r,$$

which is positive because $d(x, y_0) > r$. It is shown that $B(y_0, \delta) \subset A$: let $z \in B(y_0, \delta)$, that is, $d(y_0, z) < \delta$; it must be seen that $d(x, z) > r$. By the triangle inequality,

$$d(x, y_0) \leq d(x, z) + d(z, y_0) \implies d(x, z) \geq d(x, y_0) - d(z, y_0).$$

Since $d(z, y_0) < \delta$,

$$d(x, z) > d(x, y_0) - \delta = d(x, y_0) - (d(x, y_0) - r) = r,$$

so that $z \in A$. Hence every point of A has a ball contained in A, so that A is open. □

Exercise 3

This exercise has already been solved.

Exercise 4

This exercise has already been solved.

Exercise 5

Prove the following relations between interior, closure, unions and intersections.

i.

$$(A \cap B)^\circ = A^\circ \cap B^\circ$$

Proof. Double inclusion is proved.

(⊆). Let $x \in (A \cap B)^\circ$. By definition of interior, there is $r > 0$ with

$$B(x, r) \subset A \cap B,$$

that is, every $y \in B(x, r)$ satisfies $y \in A$ and $y \in B$. Hence $B(x, r) \subset A$, whence $x \in A^\circ$, and $B(x, r) \subset B$, whence $x \in B^\circ$; thus $x \in A^\circ \cap B^\circ$.

(⊇). Let $x \in A^\circ \cap B^\circ$. There are $r_1, r_2 > 0$ with $B(x, r_1) \subset A$ and $B(x, r_2) \subset B$. With $r = \min\{r_1, r_2\}$, the ball $B(x, r)$ is contained in both, so that $B(x, r) \subset A \cap B$ and $x \in (A \cap B)^\circ$. □

ii.

$$A^\circ \cup B^\circ \subset (A \cup B)^\circ$$

Proof. Let $x \in A^\circ \cup B^\circ$, that is, $x \in A^\circ$ or $x \in B^\circ$. If $x \in A^\circ$, there is $r > 0$ with $B(x,r) \subset A \subset A \cup B$, so that $x \in (A \cup B)^\circ$; the case $x \in B^\circ$ is analogous. In either case $x \in (A \cup B)^\circ$. □

The inclusion may be strict. In the usual $\mathbb{R}$, with $A = [0,1]$ and $B = [1,2]$ one has $A^\circ = (0,1)$, $B^\circ = (1,2)$, so that $A^\circ \cup B^\circ = (0,1) \cup (1,2)$; on the other hand $A \cup B = [0,2]$ has interior $(0,2)$. The point 1 is in $(A \cup B)^\circ$ but not in the union of the interiors.

iii.

$$\overline{A \cup B} = \overline{A} \cup \overline{B}$$

Proof. Double inclusion is proved.

($\supseteq$). Let $x \in \overline{A} \cup \overline{B}$. If $x \in \overline{A}$, for every $r > 0$ one has $B(x,r) \cap A \neq \emptyset$, and since $A \subset A \cup B$, also $B(x,r) \cap (A \cup B) \neq \emptyset$, hence $x \in \overline{A \cup B}$. The case $x \in \overline{B}$ is analogous.

($\subseteq$). By contraposition: suppose $x \notin \overline{A} \cup \overline{B}$; it must be seen that $x \notin \overline{A \cup B}$. There are $r_1, r_2 > 0$ with $B(x,r_1) \cap A = \emptyset$ and $B(x,r_2) \cap B = \emptyset$. With $r = \min\{r_1, r_2\}$, the ball $B(x,r)$ is contained in both, so that $B(x,r) \cap A = \emptyset$ and $B(x,r) \cap B = \emptyset$; therefore

$$B(x,r) \cap (A \cup B) = (B(x,r) \cap A) \cup (B(x,r) \cap B) = \emptyset,$$

so that $x \notin \overline{A \cup B}$. □

iv.

$$\overline{A \cap B} \subset \overline{A} \cap \overline{B}$$

Proof. Let $x \in \overline{A \cap B}$. For every $r > 0$, $B(x,r) \cap (A \cap B) \neq \emptyset$; let z be in that intersection (it depends on r), so that $z \in B(x,r)$, $z \in A$ and $z \in B$. Then $B(x,r) \cap A \neq \emptyset$ and $B(x,r) \cap B \neq \emptyset$ for every r, hence $x \in \overline{A}$ and $x \in \overline{B}$, that is, $x \in \overline{A} \cap \overline{B}$. □

The inclusion may be strict. In the usual $\mathbb{R}$, with $A = (0,1)$ and $B = (1,2)$ one has $A \cap B = \emptyset$, so that $\overline{A \cap B} = \emptyset$; on the other hand $\overline{A} = [0,1]$ and $\overline{B} = [1,2]$, with $\overline{A} \cap \overline{B} = \{1\} \neq \emptyset$.

Exercise 6

This exercise has already been solved.

Exercise 7

This exercise has already been solved.

Exercise 8

This exercise has already been solved.

Exercise 9

Prove that if $A \subset B$, then the diameter of A is less than or equal to the diameter of B, that is, $d(A) \leq d(B)$.

Proof. The diameter of a nonempty set is $d(A) = \sup\{d(x,y) : x, y \in A\}$. Consider the sets of distances

$$S_A = \{d(x,y) : x, y \in A\}, \qquad S_B = \{d(u,v) : u, v \in B\}.$$

Let $r \in S_A$; there are $x_0, y_0 \in A$ with $r = d(x_0, y_0)$. Since $A \subset B$, also $x_0, y_0 \in B$, so that $r = d(x_0, y_0) \in S_B$. Hence $S_A \subseteq S_B$, and since the supremum is monotone with respect to inclusion,

$$d(A) = \sup(S_A) \leq \sup(S_B) = d(B).$$

□

Exercise 10

Prove that the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is not a countable union of closed sets.

Proof. Argue by contradiction: suppose that the irrationals $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ are written as a countable union of closed sets,

$$\mathbb{I} = \bigcup_{n=1}^{\infty} F_n,$$

with each F_n closed in $\mathbb{R}$. Each $F_n \subset \mathbb{I}$, and since the irrationals do not contain any open interval (the rationals are dense), the interior of each F_n is empty, $(F_n)^\circ = \emptyset$.

On the other hand, $\mathbb{Q}$ is countable, $\mathbb{Q} = \bigcup_{k=1}^{\infty}\{q_k\}$, and each $\{q_k\}$ is closed with empty interior. Then

$$\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q} = \left(\bigcup_n F_n\right) \cup \left(\bigcup_k \{q_k\}\right)$$

is written as a countable union of closed sets with empty interior, that is, $\mathbb{R}$ would be of the first category. But $\mathbb{R}$ with the usual metric is complete, so that by Baire's Theorem 3.3 it is of the second category, a contradiction. Hence the irrationals are not a countable union of closed sets. □

Exercise 11

Given $f \in C([0,1])$, the iterated integrals f_n are defined. Prove that if for each $x \in [0,1]$ there is an $n \in \mathbb{N}$ such that $f_n(x) = 0$, then $f = 0$.

Proof. Let $Z = \{x \in [0,1] : f(x) = 0\}$ be the zero set of f. Since f is continuous, Z is closed (this is seen in the next section); it must be seen that $Z = [0,1]$. Let $U = Z^\circ$ be the largest open set where f vanishes; it is proved that U is dense in $[0,1]$.

Suppose, by contradiction, that U is not dense: there is a nonempty open interval $(a,b) \subset [0,1]$ with $(a,b) \cap U = \emptyset$. The closed interval $K = [a,b]$ is a complete subspace (closed in a complete space). By hypothesis, for each $x \in K$ there is n with $f_n(x) = 0$, so that with

$$F_n = \{x \in K : f_n(x) = 0\}$$

one has $K = \bigcup_{n=1}^{\infty} F_n$. Since K is complete, by Baire's Theorem 3.3 some F_N has nonempty interior relative to K: there is $(c, d) \subset (a, b)$ with $f_N \equiv 0$ on (c, d). Differentiating N times (using $f_k' = f_{k-1}$),

$$f_N(x) = 0 \implies f_{N-1}(x) = 0 \implies \cdots \implies f(x) = 0 \quad \forall x \in (c, d),$$

so that $(c, d) \subset Z^\circ = U$. But $(c, d) \subset (a, b)$, which contradicts $(a, b) \cap U = \emptyset$.

Hence U is dense in $[0, 1]$. Since f is continuous and vanishes on the dense set U, it vanishes on the whole space: $f(x) = 0$ for every $x \in [0, 1]$. □

4 Compactness, continuity and convergence

4.1 Preliminary Definitions

Definition 4.1 – Open cover

A family $\{U_i\}_{i\in I}$ of open sets is an open cover of $K \subset X$ if $K \subseteq \bigcup_i U_i$.

Definition 4.2 – Finite Subcover

A cover $\{U_i\}$ has a finite subcover if there are $i_1, \ldots, i_n$ with $K \subseteq U_{i_1} \cup \cdots \cup U_{i_n}$.

4.2 Definition of a Compact Set

Definition 4.3 – Compactness

K is compact if every open cover admits a finite subcover.

4.3 Properties in Metric Spaces

Proposition 4.1 – Compact Implies Closed

Every compact set K of a metric space is closed.

Proof. It is shown that K^c is open. Let $x \notin K$. For each $r > 0$ define

$$U_r = \{y \in X : d(x, y) > r\},$$

which is open. For any $z \in K$, $d(x, z) > 0$, so that $z \in U_{d(x,z)/2}$. Hence $\{U_r\}_{r>0}$ is an open cover of K.

By compactness, there is a finite subcover $U_{r_1}, \ldots, U_{r_n}$. Let $\bar{r} = \min_j r_j > 0$. The U_{r_j} are nested (the smaller r, the larger U_r), so that $\bigcup_j U_{r_j} = U_{\bar{r}}$, and $K \subseteq U_{\bar{r}}$. That is, $K \cap \overline{B(x, \bar{r})} = \emptyset$. There is a ball around x disjoint from K. □

Proposition 4.2 – Compact Implies Bounded

Every compact set K is bounded (it lives inside a ball).

Proof. Let $x_0 \in K$. The family $\{B(x_0, n)\}_{n\in\mathbb{N}}$ is an open cover of K (because every point is at finite distance from x_0). By compactness there is a finite subcover; taking N the maximum of the radii, $K \subseteq B(x_0, N)$. □

4.4 Relation to Closed Sets

Proposition 4.3 – Closed inside Compact

If K is compact and $F \subset K$ is closed, F is compact.

Proof. Given an open cover $\{U_i\}$ of F, adding the open set F^c yields a cover of K. By compactness a finite one $U_{i_1}, \ldots, U_{i_n}, F^c$ is extracted; since $F \cap F^c = \emptyset$, the U_{i_j} suffice to cover F. □

Remark. In $\mathbb{R}^n$ with the usual metric the converse holds: compact $\iff$ closed and bounded. In general it does not. For example, an infinite set with the discrete metric is closed and bounded (diameter 1) but the cover by balls of radius 1/2 (each one is a point) does not admit a finite subcover.

4.5 Sequential Compactness

Definition 4.4 – Sequential Compactness

K is sequentially compact if every sequence in K admits a subsequence converging to a point of K.

4.6 Totally Bounded Sets

Definition 4.5 – Totally Bounded

K is totally bounded if for every $\epsilon > 0$ there are $x_1, \ldots, x_n \in K$ with $K \subseteq \bigcup_i B(x_i, \epsilon)$. These centers form an ϵ-net.

4.6.1 Relation between Bounded and Totally Bounded

Proposition 4.4 – Totally Bounded Implies Bounded

Totally bounded $\implies$ bounded.

Proof. Take $\epsilon = 1$ and centers $x_1, \ldots, x_n$. Let $M = \max_i d(x_1, x_i)$. For $z \in K$ there is k with $d(z, x_k) < 1$, so that $d(z, x_1) \leq d(z, x_k) + d(x_k, x_1) < 1 + M$. Hence $K \subseteq B(x_1, M + 1)$. □

Proposition 4.5 – Sequentially Compact Implies Totally Bounded

Sequentially compact $\implies$ totally bounded.

Proof. Argue by contradiction: if it were not totally bounded, there would be $\epsilon > 0$ such that no finite ϵ-net covers K. Construct (x_n) as follows: $x_1 \in K$ arbitrary; given $x_1, \ldots, x_n$, since the balls $B(x_i, \epsilon)$ do not cover K, take x_{n+1} outside them. By construction $d(x_n, x_m) \geq \epsilon$ for $n \neq m$, so that no subsequence of (x_n) can be Cauchy, and then none converges. This contradicts sequential compactness. □

4.7 Lebesgue Number Lemma

Proposition 4.6 – Lebesgue's Lemma

If K is sequentially compact and $\{U_i\}$ is an open cover of K, there is $\eta > 0$ (the Lebesgue number) such that every ball $B(x, \eta)$ with $x \in K$ fits in some U_i.

Proof. Argue by contradiction: if no such η exists, for each n there is $x_n \in K$ with $B(x_n, 1/n)$ not contained in any U_i. By sequential compactness (x_n) has a subsequence $x_{n_k} \to x \in K$. Since $\{U_i\}$ covers K, there is U_{i_0} with $x \in U_{i_0}$, and there is $r > 0$ with $B(x, r) \subseteq U_{i_0}$. Choosing k large with $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, by the triangle inequality $B(x_{n_k}, 1/n_k) \subseteq B(x, r) \subseteq U_{i_0}$. Contradiction. □

4.8 Equivalence of Compactness

Theorem 4.1 – Equivalence Theorem

K is compact if and only if it is sequentially compact.

Proof. Both implications are proved.

(⇒) Let $(x_n) \subset K$. If the set of values $S = \{x_n\}$ is finite, some value is repeated infinitely often and a constant subsequence is obtained. If S is infinite, it is shown that it has an accumulation point in K (which then gives the subsequence by the characterization 3.9). Argue by contradiction: if it did not have one, each $z \in K$ would have $r_z > 0$ with $B(z, r_z) \cap S \subseteq \{z\}$. The family $\{B(z, r_z)\}_{z \in K}$ covers K; by compactness a finite subcover $B(z_1, r_{z_1}), \ldots, B(z_m, r_{z_m})$ is extracted. Then S is covered and has at most m elements, a contradiction.

(⇐) Let $\{U_i\}$ be an open cover. By Lebesgue there is $\eta > 0$ such that every ball of radius η fits in some U_i; being totally bounded, K is covered by finitely many $B(x_j, \eta)$, $j = 1, \ldots, n$. Take i_j with $B(x_j, \eta) \subseteq U_{i_j}$, and $K \subseteq \bigcup_j U_{i_j}$. □

4.9 Metric Characterization of Compactness

Theorem 4.2 – Compactness, Total Boundedness and Completeness

K is compact if and only if it is totally bounded and complete.

Proof. Both implications are proved.

(⇒) It was already seen that sequentially compact $\implies$ totally bounded. For completeness, a Cauchy sequence in K has a convergent subsequence (sequential compactness), and a Cauchy sequence with a convergent subsequence converges to the same limit.

(⇐) Let $(x_n) \subset K$; a convergent subsequence is sought. Diagonalization argument:

- *Step 1.* K is covered by finitely many balls of radius 1. One of them, B_1, contains infinitely many terms of (x_n). Let $(x_n^{(1)})$ be that subsequence.
- *Step k.* The sequence $(x_n^{(k-1)})$ is already available. Since K is covered by finitely many balls of radius $1/k$, some B_k contains infinitely many terms of $(x_n^{(k-1)})$; let $(x_n^{(k)})$ be the corresponding subsequence.

Define the *diagonal* $y_k = x_k^{(k)}$. It is a subsequence of (x_n) (the indices are increasing because each $(x_n^{(k)})$ is a subsequence of the previous one).

(y_k) is Cauchy: for $j, \ell \geq N$, y_j and y_ℓ are both terms of $(x_n^{(N)})$ (because the subsequences are nested), and therefore they live in B_N, which has radius $1/N$. Thus $d(y_j, y_\ell) \leq 2/N$. By completeness, (y_k) converges to a point of K. □

4.10 Heine-Borel Theorem

Theorem 4.3 – Heine-Borel

$K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. Both implications are proved.

(⇒) Already seen in general.

(⇐) If K is bounded, $K \subset [-M, M]^n$ for some M. Since closed in compact is compact (4.3), it suffices to see that $C = [-M, M]^n$ is compact. It is complete (closed in $\mathbb{R}^n$, which is complete) and totally bounded: given $\epsilon > 0$, in each coordinate $[-M, M]$ there is a finite ϵ/n-net $\{a_1, \ldots, a_p\}$; the product $\{a_1, \ldots, a_p\}^n$ is an ϵ-net for C (changing coordinate by coordinate with the triangle inequality, $d(x, a) \leq \sum_i |x_i - a_{k_i}| < \epsilon$). □

4.11 Continuity in Metric Spaces

Definition 4.6 – Continuous Function

$f : X \to Y$ is **continuous** at $a \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $f(B_X(a, \delta)) \subseteq B_Y(f(a), \epsilon)$. f is continuous if it is so at every point.

4.12 Equivalent Forms of Continuity

Theorem 4.4 – Characterizations of Continuity

For $f : X \to Y$ the following are equivalent:

i) f is continuous.
ii) $f^{-1}(U)$ is open in X for every U open in Y.
iii) $x_n \to x \implies f(x_n) \to f(x)$.

Proof. The chain $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$ is proved.

(i) ⇒ (ii): Let U be open and $a \in f^{-1}(U)$. Since U is open, there is $\epsilon > 0$ with $B(f(a), \epsilon) \subset U$; by continuity there is δ with $f(B(a, \delta)) \subset B(f(a), \epsilon) \subset U$, so that $B(a, \delta) \subset f^{-1}(U)$.

(ii) ⇒ (iii): If $x_n \to x$ and $\epsilon > 0$, $V = B(f(x), \epsilon)$ is open and $f^{-1}(V)$ is open with $x \in f^{-1}(V)$. There is $r > 0$ with $B(x, r) \subset f^{-1}(V)$; for large n, $x_n \in B(x, r)$ and $f(x_n) \in V$.

(iii) ⇒ (i): By contraposition. If f is not continuous at a, there is $\epsilon > 0$ such that for each $\delta = 1/n$ there is x_n with $d(x_n, a) < 1/n$ but $d(f(x_n), f(a)) \geq \epsilon$. Then $x_n \to a$ and $f(x_n) \not\to f(a)$. □

4.13 Continuous Functions and Compactness

Theorem 4.5 – Preservation of Compactness

If $f : X \to Y$ is continuous and $K \subset X$ is compact, $f(K)$ is compact.

Proof. Given an open cover $\{U_i\}$ of $f(K)$, the preimages $V_i = f^{-1}(U_i)$ are open and cover K. By compactness, there is a finite subcover $V_{i_1}, \ldots, V_{i_n}$; then $U_{i_1}, \ldots, U_{i_n}$ cover $f(K)$. □

4.14 Weierstrass Theorem

Corollary 4.1 – Existence of Maximum and Minimum

If K is compact and $f : K \to \mathbb{R}$ is continuous, f attains a maximum and a minimum on K.

Proof. $f(K)$ is compact in $\mathbb{R}$, so that it is closed and bounded (Heine-Borel 4.3). Closed and bounded implies that the supremum and infimum exist and are in $f(K)$ (a closed set contains its limit points, in particular its extremes). Hence there are $x_{\max}, x_{\min} \in K$ with $f(x_{\max}) = \sup$ and $f(x_{\min}) = \inf$. □

4.15 Uniform Continuity

Ordinary continuity is local (at a point a). Uniform continuity is global: it requires a uniform control over the whole domain at once.

4.15.1 Motivation: Dependence on the Point

$f(x) = 1/x$ on $(0, \infty)$ is continuous at every point. Given $a > 0$ and $\epsilon > 0$,

$$\left|\frac{1}{x} - \frac{1}{a}\right| = \frac{|x-a|}{xa}.$$

Bounding $x > a/2$, one gets $\frac{|x-a|}{xa} < \frac{2}{a^2}|x-a|$, so that $\delta \approx a^2\epsilon/2$.

The δ depends on a: as $a \to 0$, δ becomes arbitrarily small. There is no single δ that works on the whole domain.

Consequence. This lack of uniformity breaks the preservation of Cauchy sequences. The sequence $x_n = 1/n$ is Cauchy in $(0, \infty)$ but $f(x_n) = n$ is not.

4.16 Definition of Uniform Continuity

Definition 4.7 – Uniform Continuity

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ such that for all $x, y \in X$,

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon.$$

The difference lies in the order of the quantifiers: in pointwise continuity, δ is chosen after fixing x; in uniform continuity, δ is chosen beforehand and works for all points.

Example 4.1 – Lipschitz Functions

$f(x) = mx + b$ is uniformly continuous: $|f(x) - f(y)| = |m||x - y|$, so that $\delta = \epsilon/|m|$ suffices for every pair of points.

4.17 Heine-Cantor Theorem

Theorem 4.6 – Continuity on Compact Sets

If X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous.

Proof. Let $\epsilon > 0$. By continuity, for each $x \in X$ there is $\delta_x > 0$ with $f(B(x, 2\delta_x)) \subseteq B(f(x), \epsilon/2)$ (a doubled radius is taken in order to use the triangle inequality later). The $B(x, \delta_x)$ cover X; by compactness there is a finite subcover $B(x_1, \delta_{x_1}), \dots, B(x_n, \delta_{x_n})$. Take $\delta = \min_k \delta_{x_k} > 0$.

If $d(x, y) < \delta$, there is k with $x \in B(x_k, \delta_{x_k})$, and by the triangle inequality $d(y, x_k) < \delta + \delta_{x_k} \leq 2\delta_{x_k}$. Both are in $B(x_k, 2\delta_{x_k})$, so that $f(x), f(y) \in B(f(x_k), \epsilon/2)$ and

$$d(f(x), f(y)) \leq d(f(x), f(x_k)) + d(f(x_k), f(y)) < \epsilon.$$

□

4.18 Basic properties

Proposition 4.7 – Relation to Pointwise Continuity

Uniformly continuous $\implies$ continuous at every point.

Proof. Fixing $y = a$ in the definition of uniform continuity recovers continuity at a. □

Proposition 4.8 – Preservation of Cauchy Sequences

If f is uniformly continuous and (x_n) is Cauchy, $(f(x_n))$ is also Cauchy.

Proof. Given $\epsilon > 0$, take δ from uniform continuity and n_0 such that $d(x_n, x_m) < \delta$ for $n, m \geq n_0$. Then $d(f(x_n), f(x_m)) < \epsilon$. □

4.19 Contractions

Definition 4.8 – Contraction

$f : X \to X$ is a **contraction** if there is $\lambda \in [0, 1)$ with $d(f(x), f(y)) \leq \lambda\, d(x, y)$ for all x, y.

4.20 Banach Fixed-Point Theorem

Theorem 4.7 – Banach Fixed Point

Let X be a complete metric space and $f : X \to X$ a contraction. Then f has a unique fixed point, and for any $x_0 \in X$ the iteration $x_{n+1} = f(x_n)$ converges to it.

Proof. *Existence.* With x_0 fixed, let $x_{n+1} = f(x_n)$. By induction,

$$d(x_{n+1}, x_n) \leq \lambda^n d(x_1, x_0).$$

By the triangle inequality and the geometric sum,

$$d(x_{n+p}, x_n) \leq \sum_{j=0}^{p-1} \lambda^{n+j} d(x_1, x_0) \leq \frac{\lambda^n}{1-\lambda} d(x_1, x_0) \xrightarrow{n\to\infty} 0.$$

Hence (x_n) is Cauchy. By completeness it converges to some x. Since f is continuous (Lipschitz), $f(x) = \lim f(x_n) = \lim x_{n+1} = x$.

Uniqueness. If $f(x) = x$ and $f(y) = y$, $d(x, y) = d(f(x), f(y)) \leq \lambda d(x, y)$. Since $1 - \lambda > 0$, $d(x, y) = 0$. □

4.21 Other classical fixed-point theorems

Theorem 4.8 – Brouwer's Theorem

Every continuous map from a closed ball of a Euclidean space into itself admits a fixed point.

Theorem 4.9 – Schauder's Theorem

Every continuous map from a nonempty compact convex set K of a Banach space into K admits a fixed point.

Theorem 4.10 – Kakutani's Theorem

It generalizes Brouwer to functions that return sets instead of points (correspondences). If $F : K \to K$ assigns to each point a closed convex subset, and satisfies a certain continuity property (upper hemicontinuity), then there is a point x such that $x \in F(x)$.

4.22 Failure of Brouwer in Infinite Dimension

One might think that Brouwer's theorem holds in any normed vector space, that is, that every continuous function from the closed unit ball (convex, closed and bounded) into itself has a fixed point. This is false in infinite dimension.

The problem lies in the fact that, in infinite dimension, the closed and bounded ball is not compact.

Example 4.2 – Counterexample in the Sequence Space ℓ_2

Consider the space of square-summable sequences ℓ_2, the closed unit ball $\overline{B}(0,1)$ and the function $f : \overline{B}(0,1) \to \overline{B}(0,1)$ given by

$$f(x_1, x_2, x_3, \dots) = \left(\sqrt{1 - \|x\|^2}, x_1, x_2, \dots\right).$$

(i) *It is well defined.* The norm of the image is always 1:

$$\|f(x)\|^2 = (\sqrt{1 - \|x\|^2})^2 + x_1^2 + x_2^2 + \cdots = 1 - \|x\|^2 + \|x\|^2 = 1,$$

so that the image falls in the ball (in fact, on the sphere).

(ii) *It is continuous.* Its components are.

(iii) *It has no fixed point.* By contradiction: suppose $f(x) = x$. Then $\|x\| = \|f(x)\| = 1$, and equating the first coordinate, $x_1 = \sqrt{1 - \|x\|^2} = 0$; equating the second, $x_2 = x_1 = 0$, and inductively $x_n = 0$ for every n. Hence $x = 0$, whose norm is 0, which contradicts $\|x\| = 1$. Therefore there is no fixed point: f moves every point.

4.23 Homeomorphisms

Definition 4.9 – Homeomorphism

A function $f : X \to Y$ is a **homeomorphism** if it satisfies three conditions:

(i) f is bijective.

(ii) f is continuous.

(iii) Its inverse $f^{-1} : Y \to X$ is also continuous.

Proposition 4.9 – Properties of Homeomorphisms

Homeomorphisms preserve all topological properties. If f is a homeomorphism:

- U is open in $X \iff f(U)$ is open in Y.
- F is closed in $X \iff f(F)$ is closed in Y.
- K is compact in $X \iff f(K)$ is compact in Y.

Theorem 4.11 – Homeomorphism on Compact Sets

Let (X, d_X) be a compact metric space and (Y, d_Y) any metric space. If $f : X \to Y$ is a continuous bijective function, then f is a homeomorphism. (That is, $f^{-1} : Y \to X$ is automatically continuous).

Proof. Since f is already bijective and continuous, it suffices to prove that its inverse f^{-1} is continuous. A function is continuous if and only if the preimage of every closed set is closed; applied to f^{-1},

$$f^{-1} \text{ is continuous} \iff \forall F \subset X \text{ closed, } (f^{-1})^{-1}(F) \text{ is closed in } Y.$$

Since the preimage under the inverse is the direct image, $(f^{-1})^{-1}(F) = f(F)$, the goal reduces to seeing that f is closed. Let $F \subset X$ be closed:

(i) being closed in the compact X, F is compact;

(ii) since f is continuous, $f(F)$ is compact in Y;

(iii) in every metric space a compact set is closed, so that $f(F)$ is closed in Y.

Hence f maps closed sets to closed sets, f^{-1} is continuous and f is a homeomorphism. □

4.24 Types of Convergence

There are two main ways in which a sequence of functions can approach a limit function f.

Definition 4.10 – Pointwise Convergence

(f_n) **converges pointwise** to f if for each fixed $x \in K$ the numerical sequence $(f_n(x))$ converges to $f(x)$.

$$\forall x \in K, \quad \lim_{n\to\infty} f_n(x) = f(x)$$

In terms of ϵ: For every $\epsilon > 0$ and every x, there is an n_0 (depending on ϵ and on x) such that $n \geq n_0 \implies |f_n(x) - f(x)| < \epsilon$.

Pointwise convergence is weak because n_0 can grow without bound as x varies. To correct this, uniform convergence is introduced.

Definition 4.11 – Uniform Convergence

(f_n) **converges uniformly** to f if the maximum distance between the functions tends to zero.

$$\lim_{n\to\infty}\left(\sup_{x\in K}|f_n(x)-f(x)|\right)=0$$

In terms of ϵ: For every $\epsilon>0$, there is an n_0 (depending only on ϵ, not on x) such that for every $n\geq n_0$ and every $x\in K$:

$$|f_n(x)-f(x)|<\epsilon$$

4.25 The Space C(K) and its Completeness

$C(K)$ is the set of continuous functions $f:K\to\mathbb{R}$ with K compact, equipped with the supremum norm $d(f,g)=\max_{x\in K}|f(x)-g(x)|$. Converging in this metric is exactly converging uniformly.

Theorem 4.12 – Completeness of $C(K)$

$(C(K),d_\infty)$ is complete: every Cauchy sequence converges uniformly to a continuous function.

Proof. Let (f_n) be Cauchy in $C(K)$.

Pointwise limit. For each x, $|f_n(x)-f_m(x)|\leq d_\infty(f_n,f_m)$ and $(f_n(x))$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, define $f(x)=\lim_n f_n(x)$.

Uniform convergence. Being Cauchy, there is N such that $|f_n(x)-f_m(x)|<\epsilon$ for every x and every $n,m\geq N$. Fixing n and letting $m\to\infty$, $|f_n(x)-f(x)|\leq\epsilon$ for every x.

Continuity of the limit. By the 3ϵ argument: given a and $\epsilon>0$, choose n large so that $|f(x)-f_n(x)|<\epsilon/3$ uniformly; then use the continuity of f_n to obtain δ with $|f_n(x)-f_n(a)|<\epsilon/3$ when $d(x,a)<\delta$. By the triangle inequality,

$$|f(x)-f(a)|\leq|f(x)-f_n(x)|+|f_n(x)-f_n(a)|+|f_n(a)-f(a)|<\epsilon.$$

□

4.26 Solved exercises

Exercise 1

Prove the implications and give counterexamples.

i. Compact $\implies$ Totally Bounded

This implication has already been proved.

Counterexample: a set can be totally bounded but not compact. Consider the open interval $(0, 1)$ with the usual metric.

- *Totally bounded.* For any ϵ it is covered by finitely many intervals of length ϵ.
- *Not compact.* The open cover $\{(1/n, 1)\}_{n\in\mathbb{N}}$ has no finite subcover (it fails because it is not complete).

ii. Totally Bounded $\implies$ Bounded

This implication has already been proved.

Counterexample: a set can be bounded but not totally bounded. Consider an infinite set X with the discrete metric.

- *Bounded.* The whole set fits in the ball $B(x, 2)$, since the maximum distance is 1.
- *Not totally bounded.* For $\epsilon = 1/2$, the balls are individual points $\{x\}$, and infinitely many would be needed to cover the set.

Exercise 2

Let $K = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\} \subset \mathbb{R}$. Prove by definition that K is compact.

Proof. Let $\{U_i\}_{i\in I}$ be an open cover of K. Since $0 \in K$, there is i_0 with $0 \in U_{i_0}$, and being open there is $r > 0$ with $(-r, r) = B(0, r) \subseteq U_{i_0}$. Since $1/n \to 0$, there is N such that $1/n < r$ for every $n \geq N$, so that the tail $\{1/n : n \geq N\}$ lies in U_{i_0}. The finitely many points $1, 1/2, \ldots, 1/(N-1)$ remain; for each $1/k$ choose an open set U_{i_k} that contains it. Then

$$\{U_{i_0}, U_{i_1}, \ldots, U_{i_{N-1}}\}$$

is a finite subcover of K. Hence K is compact. □

Exercise 3

Prove that if a metric space (or subset) is totally bounded, then it is separable. (As a corollary, every compact metric space is separable).

Proof. Let K be totally bounded; a countable dense $D \subset K$ is sought.

Construction of D. For each $n \in \mathbb{N}$, with radius $\epsilon_n = 1/n$, total boundedness gives a finite net $F_n = \{x_{n,1}, \ldots, x_{n,k_n}\} \subset K$ with

$$K \subseteq \bigcup_{j=1}^{k_n} B\big(x_{n,j}, \tfrac{1}{n}\big)\,.$$

Define $D = \bigcup_{n=1}^{\infty} F_n$.

Countability. D is a countable union of finite sets, hence it is countable.

Density. D is dense if and only if it meets every nonempty open set of K. Let U be a nonempty open set and $x \in U$, with $B(x,r) \subseteq U$. By the Archimedean Property 2.3 there is N with $1/N < r$. The net F_N covers K with balls of radius $1/N$, so that there is $z \in F_N$ with $d(x,z) < 1/N < r$, whence $z \in B(x,r) \subseteq U$. Since $z \in D \cap U$, one has $D \cap U \neq \emptyset$. Hence D is dense in K. □

Exercise 4

Given a family of closed sets (F_α) such that one of them, $F_{\alpha'}$, is compact, prove that $K = \bigcap_\alpha F_\alpha$ is compact. Can it be empty?

Proof. An arbitrary intersection of closed sets is closed, so that $K = \bigcap_\alpha F_\alpha$ is closed. Moreover $K \subseteq F_{\alpha'}$, so that K is a closed set contained in the compact set $F_{\alpha'}$; by Proposition 4.3, K is compact. □

The intersection can be empty, and the empty set is compact. For example, in $\mathbb{R}$, with $F_1 = [0,1]$ (compact) and $F_2 = [2,3]$ (closed), one has $F_1 \cap F_2 = \emptyset$.

Exercise 5

Prove that X is compact $\iff$ for every decreasing sequence (F_n) of nonempty closed sets, $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.

Proof. *($\Rightarrow$).* Argue by contradiction: suppose there is a sequence (F_n) of nonempty decreasing closed sets with $\bigcap_{n=1}^{\infty} F_n = \emptyset$. With $U_n = F_n^c$ (open), by De Morgan

$$\bigcup_{n=1}^{\infty} U_n = \left(\bigcap_{n=1}^{\infty} F_n \right)^c = X,$$

so that $\{U_n\}$ is an open cover of X. By compactness there is a finite subcover $U_{n_1}, \dots, U_{n_k}$; since (F_n) decreases, (U_n) increases, and with $N = \max\{n_1, \dots, n_k\}$ one has $\bigcup_j U_{n_j} = U_N = X$. Taking complements, $F_N = \emptyset$, which contradicts the F_n being nonempty.

($\Leftarrow$). By contraposition: suppose that X is not compact. In a metric space this gives a countable cover $\{U_n\}$ with no finite subcover. With $A_n = U_1 \cup \cdots \cup U_n$ define $F_n = X \setminus A_n$. Each F_n is closed (complement of a finite union of open sets), the sequence decreases (A_n increases, so that $F_{n+1} \subset F_n$) and it is nonempty (if some $A_n = X$, then $U_1, \dots, U_n$ would be a finite subcover). However,

$$\bigcap_{n=1}^{\infty} F_n = X \setminus \bigcup_{n=1}^{\infty} A_n = X \setminus \bigcup_{n=1}^{\infty} U_n = X \setminus X = \emptyset.$$

Thus a decreasing sequence of nonempty closed sets with empty intersection is constructed. □

Exercise 6

The distance between two sets is defined as $d(A,B) = \inf\{d(x,y) : x \in A, y \in B\}$.

i.

Proposition 4.10 – Positive Distance

If A is compact and B is closed with $A \cap B = \emptyset$, then $d(A, B) > 0$.

Proof. Argue by contradiction: suppose $d(A, B) = 0$. By definition of infimum, for each n there are $x_n \in A$ and $y_n \in B$ with $d(x_n, y_n) < 1/n$, so that $d(x_n, y_n) \to 0$. Since A is compact, it is sequentially compact (4.1), and (x_n) has a subsequence $x_{n_k} \to a \in A$. By the triangle inequality,

$$d(y_{n_k}, a) \leq d(y_{n_k}, x_{n_k}) + d(x_{n_k}, a),$$

where $d(y_{n_k}, x_{n_k}) < 1/n_k \to 0$ and $d(x_{n_k}, a) \to 0$, so that $y_{n_k} \to a$. Since B is closed, $a \in B$. Then $a \in A \cap B$, which contradicts $A \cap B = \emptyset$. Hence $d(A, B) > 0$. □

ii.

If A and B are only disjoint closed sets, it may happen that $d(A, B) = 0$. As a counterexample in $\mathbb{R}^2$, consider the hyperbola and the x-axis:

$$A = \{(x, y) : xy = 1\}, \qquad B = \{(x, y) : y = 0\}.$$

- *Closed.* Both are preimages of closed sets under continuous functions.
- *Disjoint.* In A the y coordinate is never 0 (since $xy = 1$); in B it is always 0.
- *Zero distance.* With $a_n = (n, 1/n) \in A$ and $b_n = (n, 0) \in B$,

$$d(a_n, b_n) = \sqrt{(n - n)^2 + (1/n)^2} = \frac{1}{n} \to 0,$$

  so that the infimum of the distances is 0.

Without compactness, the sets can approach each other asymptotically without touching.

Exercise 7

Given compact subsets A and B of $\mathbb{R}^n$, prove that the set $A - B = \{a - b : a \in A, b \in B\}$ is compact.

Proof. It must be seen that every sequence in $A - B$ has a subsequence converging to a point of $A - B$. Let $(z_n) \subset A - B$, with $z_n = a_n - b_n$, $a_n \in A$, $b_n \in B$. Since A is compact, there is a subsequence $a_{n_k} \to a \in A$. Since B is compact, (b_{n_k}) has in turn a subsequence $b_{n_{k_j}} \to b \in B$. By the algebra of limits in $\mathbb{R}^n$ (2.2),

$$z_{n_{k_j}} = a_{n_{k_j}} - b_{n_{k_j}} \to a - b,$$

and $a - b \in A - B$. Hence $A - B$ is sequentially compact (4.1), that is, compact. □

Exercise 8

On the set c_0 of sequences of real numbers tending to 0, define the distance $d(x, y) = \sup_n |x_n - y_n|$. Prove that the closed unit ball $\overline{B}(0, 1)$ is not sequentially compact.

Proof. A set is sequentially compact if every sequence in it admits a convergent subsequence. A sequence in $\overline{B}(0,1)$ with no convergent subsequences is constructed.

Construction. Let $(e^{(n)})$ be the sequence of canonical vectors, with $e^{(n)}$ the sequence that equals 1 at position n and 0 elsewhere.

Membership. Each $e^{(n)}$ tends to 0 (it is eventually zero), so that $e^{(n)} \in c_0$, and $\|e^{(n)}\|_\infty = \sup_k |e_k^{(n)}| = 1$, so that $e^{(n)} \in \overline{B}(0,1)$.

Distance between terms. For $n \neq m$, the difference $e^{(n)} - e^{(m)}$ equals 1 at position n, -1 at position m and 0 elsewhere, so that

$$d(e^{(n)}, e^{(m)}) = \sup_k |e_k^{(n)} - e_k^{(m)}| = 1.$$

Conclusion. By contradiction: if $(e^{(n_k)})$ converged, it would be Cauchy, and its terms would be at distance $< \epsilon$ from each other; but the distance between two distinct terms is always 1. Hence no subsequence converges, and $\overline{B}(0,1)$ is not sequentially compact. □

Exercise 9

i.

Prove that $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2$ is continuous, but there is an open set G such that $f(G)$ is not open.

Proof. $f(x) = x^2$ is continuous on $\mathbb{R}$. Consider the open set $G = (-1, 1)$; its direct image is

$$f(G) = \{x^2 : -1 < x < 1\} = [0, 1).$$

The point $y = 0 \in f(G)$ is not interior: for every $\epsilon > 0$, the ball $B(0, \epsilon) = (-\epsilon, \epsilon)$ contains negative numbers, which are not in $[0, 1)$, so that $(-\epsilon, \epsilon) \not\subset [0, 1)$. Hence $f(G)$ is not open. □

ii.

Prove that $g : \mathbb{R} \to \mathbb{R}$ such that $g(x) = \frac{x^2}{1+x^2}$ is continuous, but there is a closed set F such that $g(F)$ is not closed.

Proof. g is continuous as a quotient of polynomials with nonvanishing denominator ($1 + x^2 \geq 1$). Consider the closed set $F = [0, +\infty)$. At $x = 0$ one has $g(0) = 0$; for $x > 0$ it is strictly increasing, and

$$\lim_{x \to +\infty} \frac{x^2}{1 + x^2} = 1,$$

so that $g(F) = [0, 1)$. The sequence $y_n = g(n) = \frac{n^2}{1+n^2} \in g(F)$ satisfies $y_n \to 1$, so that 1 is an accumulation point of $[0, 1)$ but $1 \notin [0, 1)$. Since it is missing a limit point, $g(F)$ is not closed. □

Exercise 10

Let $f, g : X \to Y$ be continuous functions.

i.

Prove that $A = \{x \in X : f(x) \neq g(x)\}$ is open.

Proof. Let $a \in A$, that is, $f(a) \neq g(a)$, with $d(f(a), g(a)) = \epsilon > 0$. With $r = \epsilon/2$, the balls $B(f(a), r)$ and $B(g(a), r)$ are disjoint. By continuity, there are $\delta_1, \delta_2 > 0$ with $f(B(a, \delta_1)) \subset B(f(a), r)$ and $g(B(a, \delta_2)) \subset B(g(a), r)$. With $\delta = \min\{\delta_1, \delta_2\}$, for $x \in B(a, \delta)$ one has $f(x) \in B(f(a), r)$ and $g(x) \in B(g(a), r)$; since those balls are disjoint, $f(x) \neq g(x)$ and $x \in A$. Hence $B(a, \delta) \subset A$, and A is open. □

ii.

Deduce that $E = \{x \in X : f(x) = g(x)\}$ is closed.

Proof. The set E is the complement of A: $E = A^c$. Since A is open (previous item), its complement E is closed. □

ii. (Alternative)

Another proof that $E = \{x \in X : f(x) = g(x)\}$ is closed.

Proof. Define the auxiliary function $h : X \to \mathbb{R}$, $h(x) = f(x) - g(x)$, which is continuous as a difference of continuous functions. A point x is in E if and only if $f(x) = g(x)$, that is, $h(x) = 0$, so that $E = h^{-1}(\{0\})$. Since $\{0\}$ is closed in $\mathbb{R}$ and h is continuous, E is closed. □

iii.

Prove that if D is dense in X and $f(x) = g(x)$ for every $x \in D$, then $f = g$ on all of X.

Proof. Consider $E = \{x \in X : f(x) = g(x)\}$. By hypothesis $D \subseteq E$, and taking closures, $\overline{D} \subseteq \overline{E}$. Since D is dense, $\overline{D} = X$, and since E is closed (item ii), $\overline{E} = E$. Hence $X \subseteq E$, that is, $f(x) = g(x)$ for every $x \in X$. □

iii. (Alternative)

Prove that if D is dense in X and $f(x) = g(x)$ for every $x \in D$, then $f = g$ on all of X.

Proof. Let $x \in X$. Since D is dense, $x \in \overline{D}$, so that there is $(d_n) \subset D$ with $d_n \to x$. By hypothesis $f(d_n) = g(d_n)$ for every n, and since f and g are continuous, $f(d_n) \to f(x)$ and $g(d_n) \to g(x)$. Since the sequences $(f(d_n))$ and $(g(d_n))$ coincide term by term, their limits are equal: $f(x) = g(x)$. Since x was arbitrary, $f = g$ on X. □

Exercise 11

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and such that there are $a, b \in \mathbb{R}$ with $f(a) < 0$ and $f(b) > 0$. Prove that there is a $c \in \mathbb{R}$ such that $f(c) = 0$.

Proof. Without loss of generality, $a < b$. Define

$$A = \{x \in [a, b] : f(x) \leq 0\}.$$

Existence of the supremum. Since $f(a) < 0$, $a \in A$, so that $A \neq \emptyset$; and $A \subseteq [a, b]$, so that b is an upper bound. By the Completeness Axiom (2.1) there is $c = \sup(A)$, with $a \leq c \leq b$.

$f(c) = 0$. The other two cases are ruled out by continuity.

Case $f(c) < 0$. By continuity, there is $\delta > 0$ with $f(x) < 0$ on $(c - \delta, c + \delta)$. Since $f(b) > 0$, $c < b$, and one can choose x_0 with $c < x_0 < c + \delta$; then $f(x_0) < 0$, so that $x_0 \in A$ with $x_0 > c$, which contradicts c being an upper bound.

Case $f(c) > 0$. By continuity, there is $\delta > 0$ with $f(x) > 0$ on $(c - \delta, c + \delta)$, so that no point of $(c - \delta, c]$ is in A and every $x \in A$ satisfies $x \leq c - \delta$; then $c - \delta < c$ would be an upper bound, against c being the least.

Hence $f(c) = 0$. □

Exercise 12

Prove that being bounded, being complete and being Cauchy are not homeomorphic properties (that is, they are not necessarily preserved by homeomorphisms).

By counterexamples. Consider two classical metric spaces with the usual metric:

- $X = (0, 1)$.
- $Y = \mathbb{R}$.

There is a homeomorphism between them, for example the scaled tangent:

$$f : (0, 1) \to \mathbb{R}, \quad f(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right)$$

This function is bijective, continuous and its inverse is continuous. However, the metric properties are not preserved.

i

- The space $X = (0, 1)$ is bounded (its diameter is 1).
- Its homeomorphic image $Y = \mathbb{R}$ is unbounded.

Hence, boundedness is not a topological invariant.

ii.

- The space $Y = \mathbb{R}$ is complete (every Cauchy sequence converges).
- The space $X = (0, 1)$ is not complete. *Justification:* The sequence $x_n = \frac{1}{n+1}$ is contained in $(0, 1)$. It is a Cauchy sequence (because in $\mathbb{R}$ it converges to 0, so that the terms get close together). However, its limit (0) does not belong to the space $(0, 1)$. Therefore, within X, the sequence does not converge.

Since $X \cong Y$ but one is complete and the other is not, completeness is not a topological invariant.

iii.

A homeomorphism can transform a Cauchy sequence into one that is not. Consider the homeomorphism $g : (0, 1) \to (1, +\infty)$ given by $g(x) = \frac{1}{x}$.

- The sequence $x_n = \frac{1}{n}$ (for $n \geq 2$) is in $(0, 1)$ and is Cauchy, since $\left|\frac{1}{n} - \frac{1}{m}\right| \to 0$.
- Applying g, $y_n = g(x_n) = n$.

- The image sequence $(2, 3, 4, \dots)$ is not Cauchy, since the distance between consecutive terms is always 1.

Hence, being a Cauchy sequence is a property that depends on the specific metric and is not preserved under homeomorphisms.

Exercise 13

This exercise has already been solved.

Exercise 14

Prove uniform continuity in the following cases.

i.

$f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 2x$.

Proof. It is Lipschitz: $|f(x) - f(y)| = 2|x - y|$. Given $\epsilon > 0$, with $\delta = \epsilon/2$, if $|x - y| < \delta$ then $|f(x) - f(y)| = 2|x - y| < \epsilon$. Hence f is uniformly continuous. □

ii.

$f : [0, \infty) \to \mathbb{R}$ such that $f(x) = \sqrt{x}$.

Proof. □

iii.

$f : [0, 1] \to \mathbb{R}$ such that $f(x) = x^2$.

Proof. f is continuous (polynomial) and the domain $[0, 1]$ is closed and bounded, hence compact in $\mathbb{R}$. By the Heine-Cantor Theorem 4.6, f is uniformly continuous. □

Counterexample:

Prove that $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2$ is not uniformly continuous.

Proof. By the characterization with sequences: if f were uniformly continuous, it would send sequences with $|x_n - y_n| \to 0$ to sequences with $|f(x_n) - f(y_n)| \to 0$. With $x_n = n$ and $y_n = n + 1/n$,

$$|x_n - y_n| = \frac{1}{n} \to 0,$$

but

$$|f(x_n) - f(y_n)| = \left|n^2 - \left(n + \tfrac{1}{n}\right)^2\right| = \left|-2 - \tfrac{1}{n^2}\right| = 2 + \frac{1}{n^2} \to 2 \neq 0.$$

Since the distance between the images does not tend to 0, f is not uniformly continuous. □

Exercise 15

Prove that if $f : \mathbb{R} \to \mathbb{R}$ is continuous and $\lim_{|x|\to\infty} f(x) = 0$, then f is uniformly continuous.

Proof. Given $\epsilon > 0$, a δ valid on all of $\mathbb{R}$ must be found.

Tails. Since $\lim_{|x|\to\infty} f(x) = 0$, there is $M > 0$ such that $|f(x)| < \epsilon/2$ for $|x| > M$.

Center. The interval $K = [-M-1, M+1]$ is compact, so that by the Heine-Cantor Theorem 4.6 f is uniformly continuous on K: there is $\delta_K > 0$ such that $|x-y| < \delta_K \implies |f(x) - f(y)| < \epsilon$ for $x, y \in K$.

Choice of δ. Take $\delta = \min(1, \delta_K)$, and let $x, y \in \mathbb{R}$ with $|x-y| < \delta$.

- If $|x| > M$ and $|y| > M$, then $|f(x) - f(y)| \leq |f(x)| + |f(y)| < \epsilon$.
- If, for example, $x \in [-M, M]$, since $|x-y| < \delta \leq 1$ one has $|y| \leq |x| + 1 \leq M + 1$, so that $x, y \in K$ and, by uniform continuity on K, $|f(x) - f(y)| < \epsilon$.

In all cases $|f(x) - f(y)| < \epsilon$, hence f is uniformly continuous on $\mathbb{R}$. □

5 Classification of Spaces and Convexity

Definition 5.1 – Hilbert Space

A **Hilbert** space is a vector space H with an inner product $\langle \cdot, \cdot \rangle$, complete with respect to the metric induced by $\|x\| = \sqrt{\langle x, x \rangle}$.

Definition 5.2 – Banach Space

A **Banach** space is a normed space $(X, \|\cdot\|)$ complete with respect to $d(x, y) = \|x - y\|$.

Every Hilbert space is a Banach space, but not conversely. The following are classical examples of Banach spaces that are not Hilbert spaces.

Example 5.1 – The Space ℓ^1

$\ell^1 = \{x = (x_n) : \sum |x_n| < \infty\}$, with norm $\|x\|_1 = \sum |x_n|$.

Example 5.2 – The Space ℓ^∞

$\ell^\infty = \{x = (x_n) : \sup_n |x_n| < \infty\}$, with norm $\|x\|_\infty = \sup_n |x_n|$.

Definition 5.3 – Fréchet Space

A locally convex, complete topological vector space whose topology is induced by a translation-invariant metric (typically by a countable family of seminorms).

Definition 5.4 – Locally Convex Space

A topological space whose topology is generated by a family of seminorms; or, equivalently, where the origin has a base of convex neighborhoods.

5.1 Difference between Hilbert and Banach: Parallelogram Law

Theorem 5.1 – Parallelogram Law

The norm of $(X, \|\cdot\|)$ comes from an inner product if and only if for all $x, y \in X$

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2).$$

Remark. ℓ^1 and ℓ^∞ do not satisfy the parallelogram law: it already fails with the canonical vectors e_1, e_2. That is why they are Banach but not Hilbert.

Remark. (Polarization identity). When the parallelogram law holds, the inner product is recovered from the norm:

$$\langle x, y \rangle = \tfrac{1}{4} \left(\|x + y\|^2 - \|x - y\|^2 \right).$$

5.2 Convexity and Continuity

Definition 5.5 – Convex Set

C is **convex** if $tx + (1-t)y \in C$ for all $x, y \in C$ and $t \in [0,1]$.

Proposition 5.1 – Convexity of the Closed Ball

$\overline{B}(0,1) = \{x : \|x\| \leq 1\}$ is convex in every normed space.

Proof. For x, y with $\|x\|, \|y\| \leq 1$ and $t \in [0,1]$, by the triangle inequality and homogeneity $\|tx + (1-t)y\| \leq t\|x\| + (1-t)\|y\| \leq 1$. □

5.2.1 Relation to the continuity of addition

The previous proof depends on the triangle inequality $\|x+y\| \leq \|x\| + \|y\|$, which is the same property that guarantees the continuity of addition $+ : X \times X \to X$. Geometrically, the convexity of the ball says that convex combinations (averages) do not escape the neighborhood. In locally convex spaces it is explicitly required that the origin have a base of convex neighborhoods so that the topology interacts well with the linear structure.

5.3 Relation between sequence spaces and compactness

Proposition 5.2 – Inclusion of ℓ^1 in ℓ^∞

$\ell^1 \subset \ell^\infty$ with $\|x\|_\infty \leq \|x\|_1$.

Proof. For each k, $|x_k| \leq \sum_n |x_n| = \|x\|_1$; taking the sup over k, $\|x\|_\infty \leq \|x\|_1$. □

Example 5.3

The unit ball $\overline{B}_{\ell^1}(0,1)$ is not sequentially compact. The canonical vectors $e^{(n)}$ (a 1 at position n, zero elsewhere) satisfy $\|e^{(n)}\|_1 = 1$. But for $n \neq m$,

$$\|e^{(n)} - e^{(m)}\|_1 = 2,$$

so that no subsequence is Cauchy and, since ℓ^1 is complete, none converges.

5.4 L^p and ℓ^p Spaces

The ℓ^1 and ℓ^∞ seen before are the two extremes of a much broader family.

Definition 5.6 – The Space L^p

Let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \leq p < \infty$. The space $L^p(\mu)$ (or simply L^p) is the set of measurable functions $f : X \to \mathbb{R}$ with

$$\int_X |f|^p \, d\mu < \infty.$$

To each one is assigned

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

Remark. $\| \cdot \|_p$ is not properly a norm but a seminorm: it satisfies $\|af\|_p = |a| \, \|f\|_p$ and the triangle inequality (Minkowski, below), but $\|f\|_p = 0$ only says that $f = 0$ almost everywhere, not everywhere. To make it a norm, one takes the quotient by the relation "equal almost everywhere".

Example 5.4 – The Space ℓ^p as a Particular Case

If $X = \mathbb{N}$ with the counting measure ($\mu(\{n\}) = 1$ for every n), the functions are sequences $a = (a_n)$ and the integral becomes a sum. This yields

$$\ell^p = \left\{ a = (a_n)_{n \in \mathbb{N}} : \sum_{n=1}^{\infty} |a_n|^p < \infty \right\}, \qquad \|a\|_p = \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{1/p}.$$

Proposition 5.3 – Inclusion between ℓ^p Spaces

If $1 \leq p_1 < p_2 \leq \infty$, then $\ell^{p_1} \subseteq \ell^{p_2}$ and $\|a\|_{p_2} \leq \|a\|_{p_1}$ for every $a \in \ell^{p_1}$.

Proof. If $a \in \ell^{p_1}$ with $\|a\|_{p_1} = 0$, then $a = 0$ and there is nothing to prove. By homogeneity one may suppose $\|a\|_{p_1} = 1$. Then $|a_n|^{p_1} \leq 1$ for every n, in particular $|a_n| \leq 1$. Since $p_2 > p_1$ and $|a_n| \leq 1$, one has $|a_n|^{p_2} \leq |a_n|^{p_1}$. Summing,

$$\|a\|_{p_2}^{p_2} = \sum |a_n|^{p_2} \leq \sum |a_n|^{p_1} = \|a\|_{p_1}^{p_1} = 1,$$

so that $\|a\|_{p_2} \leq 1 = \|a\|_{p_1}$. For $p_2 = \infty$, $\|a\|_\infty = \sup_n |a_n| \leq 1$. □

Proposition 5.4 – Inclusion in Finite Measure

If $\mu(X) < \infty$ and $1 \leq p_1 < p_2 < \infty$, then $L^{p_2}(\mu) \subseteq L^{p_1}(\mu)$.

Proof. Let $f \in L^{p_2}$. Split X according to where $|f|$ is large or small: $A = \{|f| \leq 1\}$, $A^c = \{|f| > 1\}$. On A, $|f|^{p_1} \leq 1$. On A^c, $|f|^{p_1} < |f|^{p_2}$ (because $|f| > 1$ and $p_1 < p_2$). Thus,

$$\int_X |f|^{p_1} d\mu = \int_A |f|^{p_1} d\mu + \int_{A^c} |f|^{p_1} d\mu \leq \mu(A) + \int_{A^c} |f|^{p_2} d\mu \leq \mu(X) + \|f\|_{p_2}^{p_2} < \infty.$$

□

Remark. In infinite measure the inclusion fails in both directions. On $\mathbb{R}$ with Lebesgue measure: $1/\sqrt{x} \cdot \mathbf{1}_{(0,1)} \in L^1 \setminus L^2$ (since $\int_0^1 x^{-1/2} \, dx = 2$ but $\int_0^1 x^{-1} \, dx = \infty$), and $1/x \cdot \mathbf{1}_{(1,\infty)} \in L^2 \setminus L^1$.

For sequences (counting measure, which is infinite) the inclusion goes the opposite way from L^p with finite measure: $\ell^{p_1} \subseteq \ell^{p_2}$ when $p_1 < p_2$.

5.5 Young's Inequality

Theorem 5.2 – Young's Inequality

Let $p, q \in (1, \infty)$ be conjugate exponents, that is,

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For every pair $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. If $a = 0$ or $b = 0$ the left-hand side is 0 and the inequality holds. Suppose then $a, b > 0$. The exponential function $\exp : \mathbb{R} \to \mathbb{R}$ is convex, so that for every pair of reals u, v and every $t \in [0, 1]$,

$$\exp\big(tu + (1-t)v\big) \leq t \exp(u) + (1-t)\exp(v).$$

Taking $t = 1/p$, $1 - t = 1/q$, $u = \ln(a^p)$ and $v = \ln(b^q)$ yields

$$\exp\left(\frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q)\right) \leq \frac{1}{p}\exp(\ln(a^p)) + \frac{1}{q}\exp(\ln(b^q)) = \frac{a^p}{p} + \frac{b^q}{q}.$$

On the other hand, the argument of the exponential on the left-hand side is $\ln(a) + \ln(b) = \ln(ab)$, so that $\exp(\cdot) = ab$. Combining both equalities gives the stated inequality. □

Remark. With $p = q = 2$ the inequality reduces to the inequality between the arithmetic and the geometric mean: $ab \leq \frac{a^2+b^2}{2}$.

Remark. Equality is attained if and only if $a^p = b^q$. This follows from equality in the convexity inequality of the exponential, which holds precisely when $u = v$, that is $\ln(a^p) = \ln(b^q)$.

5.6 Hölder's Inequality

Theorem 5.3 – Hölder's Inequality

Let $p, q \in (1, \infty)$ be conjugate exponents and $f \in L^p(\mu)$, $g \in L^q(\mu)$. Then $fg \in L^1(\mu)$ and

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q.$$

Proof. If $\|f\|_p = 0$ then $f = 0$ almost everywhere, so that $fg = 0$ almost everywhere and both sides vanish. Analogously if $\|g\|_q = 0$. In what follows suppose $\|f\|_p, \|g\|_q \in (0, \infty)$.

The normalized case $\|f\|_p = \|g\|_q = 1$ is treated first. Young's inequality 5.2 applied at each point $x \in X$ to the numbers $|f(x)|, |g(x)| \geq 0$ yields

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}.$$

The measurability of fg follows from that of f and g. Integrating the previous inequality,

$$\int_X |fg|\,d\mu \le \frac{1}{p}\int_X |f|^p\,d\mu + \frac{1}{q}\int_X |g|^q\,d\mu = \frac{\|f\|_p^p}{p} + \frac{\|g\|_q^q}{q} = \frac{1}{p}+\frac{1}{q} = 1,$$

which coincides with $\|f\|_p\|g\|_q = 1$.

For the general case define

$$\widetilde{f} = \frac{f}{\|f\|_p}, \qquad \widetilde{g} = \frac{g}{\|g\|_q},$$

which satisfy $\|\widetilde{f}\|_p = \|\widetilde{g}\|_q = 1$. By the normalized case, $\|\widetilde{f}\widetilde{g}\|_1 \le 1$, and multiplying by $\|f\|_p\|g\|_q$ yields $\|fg\|_1 \le \|f\|_p\|g\|_q$. The finiteness of the right-hand side guarantees $fg \in L^1$. □

Remark. In the case $p = q = 2$, Hölder reduces to the Cauchy–Schwarz inequality in L^2:

$$\left|\int_X fg\,d\mu\right| \le \int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2.$$

This is consistent with the inner product $\langle f, g\rangle = \int fg\,d\mu$ that structures L^2 as a Hilbert space.

5.7 Minkowski's Inequality

Theorem 5.4 – Minkowski's Inequality

Let $p \in [1,\infty)$ and $f, g \in L^p(\mu)$. Then $f + g \in L^p(\mu)$ and

$$\|f+g\|_p \le \|f\|_p + \|g\|_p.$$

Proof. The case $p = 1$ follows from the triangle inequality under the absolute value: $|f+g| \le |f| + |g|$, and integrating.

Suppose from now on $p > 1$. Let q be the conjugate exponent, $q = p/(p-1)$. In particular $(p-1)q = p$. For every $x \in X$,

$$|f(x)+g(x)|^p = |f(x)+g(x)|^{p-1}\cdot|f(x)+g(x)| \le |f(x)+g(x)|^{p-1}\big(|f(x)|+|g(x)|\big).$$

Integrating and separating the two summands,

$$\|f+g\|_p^p \le \int_X |f+g|^{p-1}|f|\,d\mu + \int_X |f+g|^{p-1}|g|\,d\mu.$$

Apply Hölder 5.3 to each of the two terms, with $|f+g|^{p-1}$ in L^q (which holds because $\int |f+g|^{(p-1)q} = \int |f+g|^p = \|f+g\|_p^p < \infty$) and $|f|$, $|g|$ in L^p:

$$\int_X |f+g|^{p-1}|f|\,d\mu \le \left(\int_X |f+g|^p\,d\mu\right)^{1/q}\|f\|_p = \|f+g\|_p^{p/q}\|f\|_p,$$

and analogously for g. Adding,

$$\|f+g\|_p^p \le \|f+g\|_p^{p/q}\big(\|f\|_p + \|g\|_p\big).$$

If $\|f+g\|_p = 0$ the conclusion holds. Otherwise both sides are divided by $\|f+g\|_p^{p/q}$; using $p - p/q = p(1-1/q) = p \cdot 1/p = 1$, one obtains

$$\|f+g\|_p^{p-p/q} = \|f+g\|_p,$$

whence

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p.$$

□

Remark. The Young, Hölder and Minkowski inequalities admit analogous versions for sequences, replacing the integral by a sum. Consequently, ℓ^p is a normed space for every $p \in [1, \infty)$.

5.8 Completeness of L^p: Riesz-Fischer Theorem

The proof that L^p is Banach rests on an alternative characterization of completeness in normed spaces, in terms of series.

Lemma 2. *A normed space $(E, \|\cdot\|)$ is Banach if and only if every absolutely convergent series is convergent. That is,*

$$\sum_{n=1}^{\infty} \|x_n\| < \infty \implies \sum_{n=1}^{\infty} x_n \text{ converges in } E.$$

Proof. Both implications are proved.

(⇒) Suppose E complete and $\sum \|x_n\| < \infty$. The partial sums $S_N = \sum_{n=1}^{N} x_n$ satisfy, for $N > M$,

$$\|S_N - S_M\| = \left\| \sum_{n=M+1}^{N} x_n \right\| \leq \sum_{n=M+1}^{N} \|x_n\|,$$

and the tail on the right-hand side tends to zero by convergence of $\sum \|x_n\|$. Hence (S_N) is Cauchy and, by completeness, it converges.

(⇐) Let $(x_n) \subset E$ be a Cauchy sequence: for each $k \in \mathbb{N}$, since (x_n) is Cauchy there is $n_k \in \mathbb{N}$ such that $\|x_n - x_m\| < 2^{-k}$ for all $n, m \geq n_k$; choosing the n_k strictly increasing one may suppose $n_1 < n_2 < \cdots$. In particular, $\|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}$ for every k.

Define $y_1 = x_{n_1}$ and $y_k = x_{n_k} - x_{n_{k-1}}$ for $k \geq 2$, so that $x_{n_K} = \sum_{k=1}^{K} y_k$ telescopically. The series $\sum_k \|y_k\|$ converges:

$$\sum_{k=1}^{\infty} \|y_k\| \leq \|y_1\| + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = \|y_1\| + 1 < \infty.$$

By hypothesis, the series $\sum_k y_k$ converges in E to some x, and consequently $x_{n_k} \to x$.

It remains to prove that the whole sequence converges to x. Given $\epsilon > 0$, since (x_n) is Cauchy there is $N_1 \in \mathbb{N}$ such that $\|x_n - x_m\| < \epsilon/2$ for all $n, m \geq N_1$. By convergence of the subsequence, there is K such that $\|x_{n_K} - x\| < \epsilon/2$ and $n_K \geq N_1$. For every $n \geq N_1$ one then has

$$\|x_n - x\| \leq \|x_n - x_{n_K}\| + \|x_{n_K} - x\| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

Theorem 5.5 – Riesz-Fischer Theorem

$L^p(\mu)$ is a Banach space for every $p \in [1, \infty)$.

Proof. By Lemma 2 it suffices to prove that every absolutely convergent series in L^p is convergent. Let $(f_n) \subset L^p(\mu)$ be a sequence such that

$$M := \sum_{n=1}^{\infty} \|f_n\|_p < \infty.$$

Step 1: domination by a function in L^p. For each N define

$$g_N(x) = \sum_{k=1}^{N} |f_k(x)|,$$

which is a nonnegative measurable function. By Minkowski's inequality 5.4,

$$\|g_N\|_p \le \sum_{k=1}^{N} \|f_k\|_p \le M,$$

so that $\int g_N^p \, d\mu \le M^p$.

The sequence $(g_N(x))$ is pointwise increasing, so that the limit $g(x) = \lim_N g_N(x) \in [0, \infty]$ exists, and g is measurable. By the monotone convergence theorem applied to (g_N^p),

$$\int_X g^p \, d\mu = \lim_N \int_X g_N^p \, d\mu \le M^p < \infty.$$

In particular $g \in L^p(\mu)$, and since $\int g^p \, d\mu$ is finite, $g(x) < \infty$ almost everywhere.

Step 2: definition of the sum. Let $N_0 = \{x : g(x) = \infty\}$. Since $\mu(N_0) = 0$, for $x \in X \setminus N_0$ the numerical series $\sum_k f_k(x)$ is absolutely convergent and therefore convergent in $\mathbb{R}$. Define

$$F(x) = \begin{cases} \displaystyle\sum_{k=1}^{\infty} f_k(x), & x \in X \setminus N_0, \\ 0, & x \in N_0. \end{cases}$$

The function F is measurable (almost-everywhere pointwise limit of the partial sums). Moreover, $|F(x)| \le g(x)$ for every $x \in X \setminus N_0$ (and on N_0, where $F = 0$), whence $\|F\|_p \le \|g\|_p < \infty$, that is $F \in L^p(\mu)$.

Step 3: convergence in L^p of the partial sums. Let $S_N(x) = \sum_{k=1}^{N} f_k(x)$. For $x \in X \setminus N_0$, $S_N(x) \to F(x)$, so that $|F(x) - S_N(x)|^p \to 0$ almost everywhere. The estimate

$$|F(x) - S_N(x)| \le |F(x)| + |S_N(x)| \le g(x) + g_N(x) \le 2g(x)$$

provides the domination $|F - S_N|^p \le 2^p g^p$, with $2^p g^p \in L^1(\mu)$. By the dominated convergence theorem,

$$\|F - S_N\|_p^p = \int_X |F - S_N|^p \, d\mu \xrightarrow{N \to \infty} 0,$$

or equivalently $\sum_{k=1}^{\infty} f_k = F$ in L^p. □

Remark. ℓ^p is also Banach. When $p = 2$ it is moreover Hilbert, with $\langle f, g \rangle = \int_X f g \, d\mu$ (and $\langle a, b \rangle = \sum a_n b_n$ in ℓ^2). For $p \neq 2$ the parallelogram law fails, so that the norm does not come from an inner product.

5.9 Linear Functionals and the Dual Space

In the next section continuous linear functionals on H (denoted H^*) appear repeatedly, so that it is convenient to fix the definitions. Everything that follows holds in any normed space.

Definition 5.7 – Linear functional

Let V be an $\mathbb{R}$-vector space. $\varphi : V \to \mathbb{R}$ is a linear functional if

- $\varphi(ax) = a\,\varphi(x)$ for every $a \in \mathbb{R}$, $x \in V$,
- $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in V$.

Proposition 5.5 – Continuity of a linear functional

Let V be normed and $\varphi : V \to \mathbb{R}$ linear. The following are equivalent:

(i) φ is continuous.
(ii) φ is continuous at 0.
(iii) φ is bounded: there is $M \geq 0$ with $|\varphi(x)| \leq M\|x\|$ for every x.

When they hold, the minimum M that works is

$$\|\varphi\| = \sup\{|\varphi(x)| : \|x\| \leq 1\}.$$

Proof. The chain (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1) is proved.

(1) $\Rightarrow$ (2): continuity at every point gives, in particular, continuity at 0.

(2) $\Rightarrow$ (3): Since φ is continuous at 0 and $\varphi(0) = 0$, for $\epsilon = 1$ there is $\delta > 0$ such that $\|x\| < \delta \implies |\varphi(x)| < 1$. Given $x \neq 0$, take $x' = \frac{\delta}{2\|x\|}x$, which satisfies $\|x'\| = \delta/2 < \delta$, and by linearity

$$|\varphi(x)| = \frac{2\|x\|}{\delta}|\varphi(x')| < \frac{2}{\delta}\|x\|.$$

Thus $M = 2/\delta$ works.

(3) $\Rightarrow$ (1): If $|\varphi(x)| \leq M\|x\|$, by linearity $|\varphi(x) - \varphi(y)| = |\varphi(x - y)| \leq M\|x - y\|$. That is, φ is Lipschitz (with constant M), in particular continuous at every point.

The norm. The inf of the M that work is exactly $\sup_{\|x\|\leq 1} |\varphi(x)|$: if M works, $|\varphi(x)| \leq M$ for $\|x\| \leq 1$, so that the sup is $\leq M$; conversely, if $S = \sup_{\|x\|\leq 1} |\varphi(x)|$, for any $x \neq 0$, $|\varphi(x)| = \|x\| \cdot |\varphi(x/\|x\|)| \leq S\|x\|$, so that S works. □

Definition 5.8 – Dual space

The dual of a normed space V is

$$V^* = V' = \{\varphi : V \to \mathbb{R} : \varphi \text{ is linear and continuous}\},$$

with the norm $\|\varphi\| = \sup_{\|x\|\leq 1} |\varphi(x)|$. V^* is always Banach, even if V is not.

5.10 Riesz Representation Theorem

In a Hilbert space, every continuous linear functional consists of taking the inner product against some vector. This identifies H with its own dual H^* and justifies freely exchanging functionals with elements of H in the next section.

Theorem 5.6 – Riesz Representation Theorem

Let H be a Hilbert space and $\varphi : H \to \mathbb{R}$ a continuous linear functional. There is a unique $z \in H$ such that

$$\varphi(x) = \langle x, z\rangle \quad \text{for every } x \in H,$$

and moreover $\|\varphi\|_{H^*} = \|z\|_H$. Thus, the map $z \mapsto \langle \cdot, z\rangle$ is a bijective isometry $H \to H^*$ (hence the notation $H^* \cong H$).

Proof. *Existence.* If $\varphi \equiv 0$, $z = 0$ works. Otherwise, consider the kernel

$$\mathrm{Ker}(\varphi) = \{x \in H : \varphi(x) = 0\}.$$

By continuity of φ, $\mathrm{Ker}(\varphi)$ is closed. Since $\varphi \not\equiv 0$, it is *proper* (strictly contained in H). By Corollary 6.2, $\mathrm{Ker}(\varphi)^\perp \neq \{0\}$, so that there is $\omega \in \mathrm{Ker}(\varphi)^\perp$ with $\|\omega\| = 1$.

Take $z = \varphi(\omega) \cdot \omega$: for any $x \in H$, the vector

$$x - \frac{\varphi(x)}{\varphi(\omega)}\omega$$

is in $\mathrm{Ker}(\varphi)$, so that it is orthogonal to ω:

$$0 = \left\langle x - \frac{\varphi(x)}{\varphi(\omega)}\omega, \omega \right\rangle = \langle x, \omega\rangle - \frac{\varphi(x)}{\varphi(\omega)}.$$

Solving, $\varphi(x) = \varphi(\omega)\langle x, \omega\rangle = \langle x, z\rangle$.

Uniqueness. If $\varphi(x) = \langle x, z\rangle = \langle x, z'\rangle$ for every x, then $\langle x, z - z'\rangle = 0$ for every x; taking $x = z - z'$ yields $\|z - z'\|^2 = 0$ and $z = z'$.

Isometry. By Cauchy–Schwarz, $|\varphi(x)| = |\langle x, z\rangle| \leq \|x\|\|z\|$, so that $\|\varphi\| \leq \|z\|$. And equality is attained with $x = z/\|z\|$: $\varphi(z/\|z\|) = \|z\|$. □

6 Separation theorems

Theorem 6.1 – Projection onto a Closed Convex Set

Let H be a Hilbert space and let $C \subset H$ be a closed, convex, nonempty subset. For every $x \in H$, there is a unique element $c_0 \in C$ such that

$$\|x - c_0\| = d(x, C) = \inf_{c \in C} \|x - c\|$$

That is, c_0 realizes the minimum distance from x to the set C.

Proof. *Existence.* Let $\delta = d(x, C)$ and $(c_n) \subset C$ minimizing: $\|x - c_n\| \to \delta$. It is shown that (c_n) is Cauchy.

By the parallelogram law 5.1 applied to $x - c_n$ and $x - c_m$,

$$\|c_n - c_m\|^2 = 2(\|x - c_n\|^2 + \|x - c_m\|^2) - 4\left\|x - \tfrac{c_n + c_m}{2}\right\|^2.$$

By convexity, $\frac{c_n + c_m}{2} \in C$, so that $\|x - \frac{c_n + c_m}{2}\| \geq \delta$ and

$$\|c_n - c_m\|^2 \leq 2(\|x - c_n\|^2 + \|x - c_m\|^2) - 4\delta^2 \xrightarrow{n,m \to \infty} 4\delta^2 - 4\delta^2 = 0.$$

By completeness $c_n \to c_0$, and since C is closed $c_0 \in C$. By continuity of the norm, $\|x - c_0\| = \delta$.

Uniqueness. If $c_0, c_1 \in C$ both realize the distance, the same computation gives $\|c_0 - c_1\|^2 \leq 4\delta^2 - 4\delta^2 = 0$. □

6.1 Projection onto Subspaces and Complementarity

Corollary 6.1 – Orthogonal Projection

If S is a closed subspace of H, for every $x \in H$ there is a unique $s_0 = P_S(x) \in S$ that minimizes the distance. It is the **orthogonal projection** of x onto S.

Corollary 6.2 – Nontrivial $S^\perp$

If S is a closed *proper* subspace of H, then $S^\perp \neq \{0\}$.

Proof. Take $x \in H \setminus S$. By Corollary 6.1, $x = s + s^\perp$ with $s \in S$ and $s^\perp \in S^\perp$. If $s^\perp = 0$ then $x = s \in S$, a contradiction. Thus $s^\perp \neq 0$. □

6.2 The Failure in Banach Spaces

The properties of existence, uniqueness and complementarity depend essentially on the inner product. In general Banach spaces they do not hold.

Remark. In a Banach space X:

(i) The best approximation to a closed subspace may fail to be unique (norm not strictly convex).

(ii) There may be no element that realizes the distance (nonreflexive space).

(iii) A closed subspace may fail to be complemented.

Example 6.1 – Failure of Complementarity: c_0 in ℓ^∞

Let $X = \ell^\infty$ and $S = c_0$ (sequences tending to 0). c_0 is closed in ℓ^∞, but by a theorem of Phillips (1940) it is not complemented: there is no closed W with $\ell^\infty = c_0 \oplus W$ topologically. Hilbert geometry is considerably more rigid than Banach geometry.

6.3 Strict separation theorem

Theorem 6.2 – Separation of Compact Convex Sets

Let H be a Hilbert space. Let A and B be two compact, convex, disjoint ($A \cap B = \emptyset$) subsets of H. Then there is a continuous linear functional $f \in H^*$ that strictly separates the sets:

$$\max_{a \in A} f(a) < \min_{b \in B} f(b)$$

Proof. It reduces to a single set: $K = B - A = \{b - a : a \in A, b \in B\}$.

K is convex (sum/difference of convex sets). K is compact: given $(z_n) = (b_n - a_n)$, by compactness of A take $a_{n_k} \to a \in A$; by compactness of B, $b_{n_{k_j}} \to b \in B$; then $z_{n_{k_j}} \to b - a \in K$.

Applying Theorem 6.1 with the origin against K, there is a unique $z_0 \in K$ with $\|z_0\| = d(0, K)$. Moreover $\|z_0\| > 0$ because $A \cap B = \emptyset$ implies $0 \notin K$. Let $z_0 = b_0 - a_0$ and define $f(x) = \langle x, z_0 \rangle \in H^*$.

By the property of the projection onto convex sets, $\langle k - z_0, z_0 \rangle \geq 0$ for every $k \in K$, that is, $\langle k, z_0 \rangle \geq \|z_0\|^2$. For $k = b - a$,

$$f(b) - f(a) \geq \|z_0\|^2 > 0.$$

Since A, B are compact, f attains its extremes, and

$$\min_{b \in B} f(b) \geq \max_{a \in A} f(a) + \|z_0\|^2 > \max_{a \in A} f(a).$$

□

6.4 Non-strict separation theorem

Definition 6.1 – Convex Body

$K \subset \mathbb{R}^n$ is a **convex body** if it is convex, compact and with nonempty interior.

Theorem 6.3 – Separation of Convex Bodies

Let A, B be convex bodies in $\mathbb{R}^n$ with $(A \cap B)^\circ = \emptyset$. There is a nonzero linear functional $f : \mathbb{R}^n \to \mathbb{R}$ with

$$\max_{a \in A} f(a) \leq \min_{b \in B} f(b).$$

Proof. Define the *shrunken* sets $A_m = \{x \in A : d(x, A^c) \geq 1/m\}$ and analogously B_m. They satisfy:

- *Compact.* They are the intersection of A (compact) with $g^{-1}([1/m, \infty))$ where $g(x) = d(x, A^c)$ is continuous.
- *Convex.* The characterization $x \in A_m \iff B(x, 1/m) \subseteq A$ is used. If $x, y \in A_m$ and $\|h\| < 1/m$, then $z + h = t(x + h) + (1 - t)(y + h)$ is a convex combination of points of A, hence in A.
- *Increasing and disjoint.* $A_m \subseteq A_{m+1}$ because $1/(m + 1) < 1/m$. If $x \in A_m \cap B_m$, $B(x, 1/m) \subset A \cap B$, contradicting $(A \cap B)^\circ = \emptyset$.

For each m, A_m, B_m are disjoint compact convex sets. By Theorem 6.2 there is $u_m \in S^{n-1}$ that separates them. Concretely, let

$$K_m = \left\{ u \in S^{n-1} : \max_{a \in A_m} \langle a, u \rangle \leq \min_{b \in B_m} \langle b, u \rangle \right\}.$$

It is closed in S^{n-1} (limit of $\leq$ inequalities), nonempty by strict separation, and decreasing: if $u \in K_{m+1}$ separates $A_{m+1} \supseteq A_m$ from $B_{m+1} \supseteq B_m$, in particular it separates A_m from B_m.

Since S^{n-1} is compact and the K_m are nonempty nested closed sets, $\bigcap_m K_m \neq \emptyset$. Take w there and define $f(x) = \langle x, w \rangle$.

For $x \in A^\circ$ and $y \in B^\circ$, there is m with $x \in A_m$, $y \in B_m$, so that $f(x) \leq f(y)$. Since $A = \overline{A^\circ}$ and $B = \overline{B^\circ}$ (they are convex bodies) and f is continuous, the inequality extends to A and B. □

7 Extra Exercises

Exercise 1

Statement: Given the sequences (x_n) and (y_n), the shuffled sequence (z_n) is defined as:

$$z_n = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ y_{\frac{n}{2}} & \text{if } n \text{ is even} \end{cases}$$

That is: $z = (x_1, y_1, x_2, y_2, x_3, \dots)$. Prove that (z_n) is convergent $\iff$ (x_n) and (y_n) are and have the same limit.

Proof. *($\Rightarrow$).* Suppose that (z_n) converges to L. Every subsequence of a convergent sequence converges to the same limit. The odd terms of (z_n) are $z_{2k-1} = x_k$, so that (x_n) is a subsequence of (z_n) and $x_n \to L$; the even ones are $z_{2k} = y_k$, so that $y_n \to L$. Hence both converge to the same limit L.

($\Leftarrow$). Suppose that $x_n \to L$ and $y_n \to L$; it must be proved that $z_n \to L$. Let $\epsilon > 0$. There are N_1, N_2 with $|x_n - L| < \epsilon$ for $n > N_1$ and $|y_n - L| < \epsilon$ for $n > N_2$. With $N_0 = \max(2N_1, 2N_2)$, let $k > N_0$:

- if $k = 2m-1$ is odd, $z_k = x_m$ and, since $k > 2N_1$, $m > N_1$, so that $|z_k - L| = |x_m - L| < \epsilon$;
- if $k = 2m$ is even, $z_k = y_m$ and, since $k > 2N_2$, $m > N_2$, so that $|z_k - L| = |y_m - L| < \epsilon$.

In both cases $|z_k - L| < \epsilon$ for $k > N_0$, hence $z_n \to L$. □

Exercise 2

Let $X = (\mathbb{R}^2, d)$ where the metric d is given by:

$$d((a_1, a_2), (b_1, b_2)) = \begin{cases} |a_1 - b_1| & \text{if } a_2 = b_2 \\ |a_1 - b_1| + |a_2 - b_2| + 1 & \text{if } a_2 \neq b_2 \end{cases}$$

i.

By definition, the open ball is:

$$B((0,0), 1) = \{(x, y) \in \mathbb{R}^2 : d((0,0), (x, y)) < 1\}$$

According to the value of y:

- *Case* $y = 0$. The condition of the metric is the first branch ($a_2 = b_2$).

  $$d((0,0), (x, 0)) = |x - 0| = |x|$$

  The condition of the ball is $|x| < 1$. This describes the segment $(-1, 1)$ on the X axis.

- *Case* $y \neq 0$. The condition of the metric is the second branch ($a_2 \neq b_2$).

  $$d((0,0), (x, y)) = |x - 0| + |y - 0| + 1 = |x| + |y| + 1$$

  To be in the ball, this distance must be less than 1.

  $$|x| + |y| + 1 < 1 \implies |x| + |y| < 0$$

  Since absolute values are nonnegative, this is impossible. No point with $y \neq 0$ belongs to the ball.

Hence, the open ball is the open segment on the X axis:

$$B((0,0),1) = (-1,1) \times \{0\}$$

ii.

To determine whether the vertical segment $S = \{0\} \times [0,1]$ is compact in this metric, the fact that a set is compact if every sequence in it admits a convergent subsequence is used.

Proof. Consider the sequence $z_n = (0, 1/n) \in S$. For $n \neq m$, their second coordinates differ, so that by the metric

$$d(z_n, z_m) = |0 - 0| + \left|\frac{1}{n} - \frac{1}{m}\right| + 1 \geq 1.$$

Since the distance between distinct terms is always ≥ 1, no subsequence is Cauchy and, therefore, none converges. Hence S is not compact. □

Exercise 3: Intersection of Open Sets

i.

Let $f_1, \ldots, f_n : X \to \mathbb{R}$ be continuous. Prove that $U = \{x \in X : f_i(x) > 0 \forall i\}$ is open.

Proof. Let $U_i = \{x \in X : f_i(x) > 0\} = f_i^{-1}((0, +\infty))$. Since $(0, +\infty)$ is open in $\mathbb{R}$ and f_i is continuous, each U_i is open. The set in the statement is $U = \bigcap_{i=1}^{n} U_i$, a finite intersection of open sets, hence open. □

ii.

Show that if there are infinitely many functions, the result does not hold.

Counterexample: consider $X = \mathbb{R}$ with the usual metric and the family of continuous functions $f_n(x) = \frac{1}{n} - |x|$. The set where $f_n(x) > 0$ is

$$U_n = \left\{x : |x| < \tfrac{1}{n}\right\} = \left(-\tfrac{1}{n}, \tfrac{1}{n}\right),$$

which is open. Its infinite intersection is

$$\bigcap_{n=1}^{\infty} U_n = \{0\},$$

since the only x with $|x| < 1/n$ for every n is $x = 0$. But $\{0\}$ is closed, not open, so that the property fails for infinite intersections.

Exercise 4:

A metric space X is called connected if it is not the union of two nonempty open sets with empty intersection. The space X has the property: for all $x, y \in X$, there is a continuous function $f : [0,1] \to X$ such that $f(0) = x$ and $f(1) = y$. Prove that X is connected.

Proof. Argue by contradiction: suppose that X is not connected, that is, $X = U \cup V$ with U, V nonempty disjoint open sets. Choose $x \in U$ and $y \in V$; by hypothesis there is a continuous path $f : [0,1] \to X$ with $f(0) = x$ and $f(1) = y$. Let $A = f^{-1}(U)$ and $B = f^{-1}(V)$:

(i) A and B are open in $[0,1]$ by continuity of f;

(ii) they are disjoint, since $f^{-1}(U \cap V) = f^{-1}(\emptyset) = \emptyset$;

(iii) $A \cup B = f^{-1}(U \cup V) = f^{-1}(X) = [0,1]$;

(iv) they are nonempty: $0 \in A$ (since $f(0) = x \in U$) and $1 \in B$ ($f(1) = y \in V$).

Then $[0,1]$ would be a union of two nonempty disjoint open sets, which contradicts $[0,1]$ being connected. Hence X is connected. □

Exercise 5:

Prove that if V is open, then $V \cap \overline{A} \subset \overline{V \cap A}$. (Note: this version of the statement is assumed, since the literal one $A \cap V \subset A \cap V$ always holds).

Proof. Let $x \in V \cap \overline{A}$; it must be seen that $x \in \overline{V \cap A}$, that is, that every ball centered at x meets $V \cap A$. Let $r > 0$. Since $x \in V$ and V is open, there is $\delta > 0$ with $B(x,\delta) \subset V$; with $\rho = \min(r, \delta)$, the ball $B(x,\rho)$ is contained in $B(x,r)$ and in V. Since $x \in \overline{A}$, $B(x,\rho) \cap A \neq \emptyset$; let z be in that intersection. Then $z \in B(x,r)$, $z \in V$ and $z \in A$, so that $z \in B(x,r) \cap (V \cap A)$. Since this holds for every r, one has $x \in \overline{V \cap A}$. □

Exercise 6:

Let x_n be a sequence in X such that $x_n \to x$. Prove that $K = \{x_n : n \in \mathbb{N}\} \cup \{x\}$ is compact.

Proof. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ be an open cover of K. Since $x \in K$, there is α_0 with $x \in U_{\alpha_0}$, and being open there is $r > 0$ with $B(x,r) \subseteq U_{\alpha_0}$. Since $x_n \to x$, with $\epsilon = r$ there is N such that $x_n \in B(x,r) \subseteq U_{\alpha_0}$ for every $n > N$: the tail is covered by U_{α_0}. The remaining terms $x_1, \ldots, x_N$ are finitely many; for each one choose an open set U_{α_k} with $x_k \in U_{\alpha_k}$. The finite family $\{U_{\alpha_0}, U_{\alpha_1}, \ldots, U_{\alpha_N}\}$ covers K, hence K is compact. □

Exercise 7:

Prove that if $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ are continuous, then so is $f + g$. Can the sum of two non-continuous functions be continuous? And the sum of a continuous and a non-continuous one?

i.

Proof. Let $a \in X$; by the characterization with sequences (Theorem 4.4), let (x_n) with $x_n \to a$. Since f and g are continuous at a, $f(x_n) \to f(a)$ and $g(x_n) \to g(a)$, and by the algebra of limits in $\mathbb{R}$ (2.2),

$$(f+g)(x_n) = f(x_n) + g(x_n) \to f(a) + g(a) = (f+g)(a).$$

Hence $f + g$ is continuous at a, and since a is arbitrary, on all of X. □

ii.

Yes. Consider the Dirichlet function $f(x) = 1$ if $x \in \mathbb{Q}$ and $f(x) = 0$ if $x \notin \mathbb{Q}$, discontinuous at every point, and $g = -f$, also discontinuous at every point. Their sum $(f+g)(x) = 0$ for every x is the constant function 0, which is continuous.

iii.

Proof. Suppose by contradiction that f is continuous, g discontinuous and the sum $h = f + g$ continuous. Solving $g = h - f$, since h and f are continuous their difference is too, so that g would be continuous, against the hypothesis. Hence the sum of a continuous and a discontinuous function is discontinuous. □

Exercise 8:

Let $K \subset \mathbb{R}^n$ be a compact set and $F \subset \mathbb{R}^n$ a closed set. Let $K + F = \{a + b : a \in K, b \in F\}$. Prove that $K + F$ is closed and prove that if K and F are only closed, $K + F$ may fail to be closed.

i.

Proof. Let $(z_n) \subset K + F$ with $z_n \to z$ in $\mathbb{R}^n$; it must be seen that $z \in K + F$. For each n, $z_n = k_n + f_n$ with $k_n \in K$, $f_n \in F$. Since K is compact, there is a subsequence $k_{n_j} \to k \in K$. Then

$$f_{n_j} = z_{n_j} - k_{n_j} \to z - k =: f,$$

and since $(f_{n_j}) \subset F$ and F is closed, $f \in F$. Hence $z = k + f$ with $k \in K$, $f \in F$, that is, $z \in K + F$, and the set is closed. □

ii.

Counterexample in $\mathbb{R}$: let

- $A = \{n + \frac{1}{n} : n \in \mathbb{N}, n \geq 2\}$, closed;
- $B = \{-n : n \in \mathbb{N}, n \geq 2\}$, closed.

Adding $a_n = n + 1/n$ and $b_n = -n$ yields $a_n + b_n = 1/n \in A + B$, which converges to 0. But $0 \notin A + B$: if $0 = a + b$ then $a = -b$, that is $n + 1/n = m$ an integer, impossible for $n \geq 2$. Thus $A + B$ has a limit point that does not belong to it, hence it is not closed.

Exercise 9:

Let $f : \mathbb{R} \to \mathbb{R}$ and $G_f \subset \mathbb{R}^2$ its graph. Prove that if f is continuous, then G_f is closed and prove that G_f closed does not imply that f is continuous.

i.

Proof. Let $(z_n) \subset G_f$ with $z_n = (x_n, y_n) \to z_0 = (x_0, y_0)$. Since they are in the graph, $y_n = f(x_n)$, and convergence in the plane gives $x_n \to x_0$ and $f(x_n) \to y_0$. By continuity of f, $f(x_n) \to f(x_0)$, and by uniqueness of the limit (2.6), $y_0 = f(x_0)$. Then $z_0 = (x_0, f(x_0)) \in G_f$, hence G_f is closed. □

ii.

Proof. Consider $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$, discontinuous at $x = 0$. Let $(x_n, f(x_n)) \to (a, b)$.

- *Case* $a \neq 0$. By continuity of $1/x$ away from the origin, $f(x_n) \to 1/a$, hence $(a, 1/a) \in G_f$.

- *Case* $a = 0$. If $x_n \neq 0$, then $|f(x_n)| = |1/x_n| \to \infty$ and the sequence does not converge in $\mathbb{R}^2$; the only way to converge to a point of the Y axis is for it to be eventually $(0,0)$, and $(0,0) \in G_f$.

The graph contains all its limits, so that it is closed, but f is not continuous. □

Exercise 10:

Prove that the set $F \subset \mathbb{R}^2$ is totally bounded, where:

$$F = \{(0,0)\} \cup \underbrace{\left\{\left(\frac{1}{n}, 0\right)\right\}_{n\in\mathbb{N}}}_{S_1} \cup \underbrace{\left\{\left(1 - \frac{1}{n}, \frac{1}{n}\right)\right\}_{n\in\mathbb{N}}}_{S_2}$$

Proof. Let $\epsilon > 0$; a finite cover of F by balls of radius ϵ is sought. The sequence S_1 has limit $L_1 = (0,0)$ and S_2 has limit $L_2 = (1,0)$.

(i) There are N_1, N_2 such that

$$\begin{aligned} n > N_1 &\implies \left(\tfrac{1}{n}, 0\right) \in B(L_1, \epsilon), \\ n > N_2 &\implies \left(1 - \tfrac{1}{n}, \tfrac{1}{n}\right) \in B(L_2, \epsilon), \end{aligned}$$

so that all the terms with large n lie in two balls.

(ii) The remaining points are finitely many. With $I = \{1, \ldots, \max(N_1, N_2)\}$, for each $k \in I$ take the balls

$$\mathcal{B}_k^1 = B\left(\left(\tfrac{1}{k}, 0\right), \epsilon\right), \qquad \mathcal{B}_k^2 = B\left(\left(1 - \tfrac{1}{k}, \tfrac{1}{k}\right), \epsilon\right).$$

(iii) The family

$$\mathcal{U} = \{B(L_1, \epsilon), B(L_2, \epsilon)\} \cup \bigcup_{k=1}^{\max(N_1,N_2)} \{\mathcal{B}_k^1, \mathcal{B}_k^2\}$$

covers F and consists of $2 + 2\max(N_1, N_2)$ balls. Since it is finite for every $\epsilon > 0$, F is totally bounded.

□

Exercise 11

Let A be a nonempty subset of a metric space X. Prove that:

$$\bigcap_{r>0} \bigcup_{a\in A} B(a, r) = \overline{A}$$

Proof. Double inclusion is proved.

($\supseteq$). Let x be in the set on the left: for every $\epsilon > 0$, $x \in \bigcup_{a\in A} B(a, \epsilon)$, so that there is $a \in A$ (depending on ϵ) with $d(x, a) < \epsilon$. Then $a \in B(x, \epsilon)$, so that $B(x, \epsilon) \cap A \neq \emptyset$ for every ϵ, and by definition $x \in \overline{A}$.

($\subseteq$). Let $x \in \overline{A}$. For every $r > 0$, $B(x, r) \cap A \neq \emptyset$, so that there is $a \in A$ with $d(x, a) < r$; by symmetry $x \in B(a, r)$, hence $x \in \bigcup_{a\in A} B(a, r)$. Since this holds for every $r > 0$,

$$x \in \bigcap_{r>0} \bigcup_{a\in A} B(a, r).$$

□